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School of AI & Data for Business & Society

Paris Region · Nice Côte d'Azur

Grande Ecole Programme PGE5 — AI & Data Science

Auditing the Calibration of Eligibility Scores Produced by Large Language Models in Patient–Clinical Trial Matching

Assessing the Effect of RLHF and Uncovering Mode Collapse in Open-Source Medical LLMs

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Academic Year 2025–2026

September 2026

Acknowledgements

I would like to express my sincere gratitude to my academic supervisor, Dr. Anuradha Kar, for her guidance, insightful feedback, and constant support throughout this research project. Her expertise in artificial intelligence and her rigorous methodological approach were invaluable in shaping the direction of this work.

I am equally grateful to Aivancity — School of AI & Data for Business & Society for providing the academic environment and resources that made this project possible, and to the Grande Ecole Programme faculty for the training that prepared me for such a research endeavor.

Abstract

Keywords: large language models, calibration, clinical trial matching, RLHF, medical AI, mode collapse, expected calibration error, conformal prediction

Résumé

Mots-clés : grands modèles de langage, calibration, appariement d'essais cliniques, RLHF, IA médicale, mode collapse, expected calibration error, prédiction conforme

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List of Abbreviations

Acronym	Definition
AUC	Area Under the Curve
CI	Confidence Interval
ECE	Expected Calibration Error
EHR	Electronic Health Record
GPU	Graphics Processing Unit
IC	Intervalle de Confiance (French)
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MC	Mode Collapse
ML4H	Machine Learning for Health
NEI	Not Enough Information
NF4	Normal Float 4-bit (quantization)
NLL	Negative Log-Likelihood
NLP	Natural Language Processing
PPO	Proximal Policy Optimization
QA	Question Answering
RCT	Randomized Controlled Trial
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine-Tuning
TREC	Text REtrieval Conference

Chapter 1

Introduction

1.1 Macro Context: AI in Healthcare

1.2 Micro Context: Clinical Trial Matching

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Chapter 2

State of the Art

This chapter reviews the state of the art at the intersection of three research areas that jointly frame the present study: patient–clinical trial matching, probabilistic calibration of classifiers, and the evaluation of medically specialized large language models. Rather than surveying each area in isolation, we organize the review around the specific gap that motivates this thesis — the absence of systematic calibration audits in LLM-based clinical trial matching — and use each section to build the position of this work with respect to the closest prior contributions. Section **??** closes the chapter with a comparative table and an explicit statement of the contribution.

2.1 Patient–Clinical Trial Matching

2.1.1 Clinical Context and Recruitment Bottlenecks

Clinical trial recruitment is one of the most persistent bottlenecks in translational medicine. Despite widespread interest from patients and physicians, less than 5% of adult cancer patients enroll in a clinical trial, and 80% of trials fail to meet their recruitment targets [1]. The mismatch between patients and appropriate trials is caused, in part, by the sheer volume of open studies (over 400,000 registered on ClinicalTrials.gov at the time of writing) and the complexity of their eligibility criteria, which combine numeric thresholds, medical history requirements, and drug exclusions in dozens of criteria per trial. Manual screening of a patient against this space of criteria is labor-intensive and error-prone, and automating even a fraction of this process would substantially accelerate recruitment.

Automated matching systems have thus been a topic of active research for over a decade. The field has passed through three distinct waves, each corresponding to a different technological substrate.

2.1.2 Pre-LLM Approaches: Rule-Based and Embedding Systems

The first wave of matching systems relied on structured information extraction. EliXR [2], EliIE [3], and Criteria2Query [4] transformed free-text eligibility criteria into structured OMOP or SQL queries, matched against structured electronic health records. These systems produce deterministic Boolean outputs (match or no match) and provide no probability estimates or uncertainty quantification. They perform well on criteria that map cleanly to structured concepts (age, diagnoses, laboratory values) but struggle with criteria requiring nuanced interpretation.

The second wave introduced end-to-end neural matching. DeepEnroll [5] encoded criteria with BERT and patient EHRs with hierarchical embeddings, framing matching as textual entailment and reporting PR-AUC and F1 scores. COMPOSE [6] added a cross-modal pseudo-siamese network with a composite inclusion/exclusion loss, achieving 98.0% patient–criterion AUC and 83.7% patient–trial accuracy. Neither system reported calibration metrics. This omission is not accidental: the primary evaluation targets of these systems were discrimination metrics (AUC, F1) rather than the reliability of the probabilistic outputs.

2.1.3 The LLM Wave: 2023–2026

The third and current wave dates from the release of GPT-4 and the subsequent proliferation of open-source large language models. The most influential system in this wave is TrialGPT [1], published in *Nature Communications* in 2024. TrialGPT is a three-module pipeline: a retrieval stage recovers more than 90% of relevant trials in less than 6% of the collection using a hybrid BM25 and MedCPT retriever; a matching stage emits a per-criterion ordinal label ("included / not included / no relevant information"); and a ranking stage aggregates criterion labels into trial-level scores. TrialGPT reports a criterion-level accuracy of 87.3% and correlation patterns against the ground truth, but does not report any calibration metric — no ECE, no Brier score, no reliability diagram, and no monotonicity check across the ordinal scale.

TrialGPT is not an isolated case. RECTIFIER [7], published in *NEJM AI*, applies a retrieval-augmented GPT-4 to the COPILOT-HF heart-failure trial and reports 97.9–100% accuracy with Matthews correlation coefficients between 0.837 and 1.00. The system has been deployed clinically at Mass General Brigham, yet its outputs are reported only as accuracy and MCC, without probabilistic reliability assessment. PRISM and its variant OncoLLM [8], published in *npj Digital Medicine*, introduce a weighted-tier scoring method with concept-level importance and report top-k accuracy and nDCG (68% vs. 62.6% for GPT-3.5) but again omit calibration. Trial-LLaMA distills open-source Llama on synthetic criterion labels [9] and reports accuracy alone. Zero-shot matching approaches by Wornow et al. [10] achieve 0.93 micro-F1 on the n2c2 2018 dataset and elicit a self-declared confidence per criterion — the closest any published system comes to uncertainty quantification — but the self-declared confidence is never calibrated against ground truth. More recent systems including TrialMatchAI [11], multimodal pipelines by Callies et al. [12], and cross-modal trial-site selectors extend the scope of matching but continue the pattern of accuracy-only reporting.

The definitive scoping review of the field, published by Chen et al. [13] in *JCO Clinical Cancer Informatics*, screened 2,357 papers and included 24 LLM-based matching systems, of which 21 were published in 2024. The review confirms that *none* of these systems reports calibration metrics: no ECE, no Brier score, no reliability diagram, no calibration-in-the-large, and no slope. This gap defines the primary methodological opportunity addressed by the present thesis.

2.2 Probabilistic Calibration of Classifiers

2.2.1 Foundations and the Modern Miscalibration Problem

Probability calibration — the requirement that predicted probabilities match empirical accuracies — has a long history in statistical prediction. Historical foundations include Platt scaling [14] for logistic regression outputs, isotonic regression [15] for non-parametric monotone calibration, and the systematic comparisons of Niculescu-Mizil and Caruana [16]. For decades, calibration was treated as a solved problem: shallow classifiers such as logistic regression, naïve Bayes, and calibrated support vector machines were well calibrated after appropriate post-hoc correction.

This picture changed with the rise of deep learning. Guo, Pleiss, Sun, and Weinberger [17] showed in 2017 that modern deep neural networks, despite their superior accuracy, are *systematically overconfident* relative to shallower architectures: a network with 85% accuracy may assign 99% confidence to its predictions, including its errors. This paper introduced the Expected Calibration Error (ECE) as the standard metric for measuring miscalibration in modern classifiers, and demonstrated the effectiveness of temperature scaling, a single-parameter post-hoc method that does not modify argmax predictions.

Since 2017, the calibration literature has expanded rapidly. Alternative metrics have been proposed to address specific limitations of ECE: adaptive-ECE, class-wise ECE, and the more refined estimators of Nixon et al. [18] and Kumar, Liang, and Ma [19].

The Brier score [20] provides a joint measure of calibration and sharpness. Beyond ECE, distribution-free confidence intervals for calibration have been studied by Gupta and Ramdas [21].

2.2.2 Clinical Importance of Calibration

The clinical importance of calibration has been argued particularly forcefully by Van Calster et al. [22], whose 2019 *BMC Medicine* article titled "Calibration: the Achilles heel of predictive analytics" establishes calibration as a prerequisite for clinical deployment of any predictive model. Van Calster and colleagues distinguish four progressive levels of calibration — calibration in the large, weak calibration, moderate calibration, and strong calibration — and argue that most published clinical prediction models fail even the weakest level. They also introduce the threshold ECE < 0.05 as a practical benchmark for clinically usable calibration, which the present thesis adopts as the rejection criterion for Hypothesis H1 (Section 4.2).

The 2024 update of the TRIPOD guidelines, known as TRIPOD+AI [23], formalizes calibration as a required reporting element for any AI-based clinical prediction model. Under TRIPOD+AI, a paper introducing a clinical AI model must report calibration curves, calibration-in-the-large, and calibration slopes; failure to do so is now grounds for methodological rejection by several leading medical journals. This regulatory shift makes the absence of calibration reporting in LLM-based trial matching systems increasingly untenable.

2.2.3 Recalibration Methods

Post-hoc recalibration methods apply a monotone transformation to uncorrected confidences after the classifier has been trained. The three methods evaluated in the present thesis — temperature scaling, isotonic regression, and Platt scaling — represent the classical choices; more elaborate methods include beta calibration [24], Dirichlet calibration [25], and the matrix/vector scaling family. Intrinsic methods, which modify the training objective itself, include label smoothing [26] and focal loss [27]. Distribution-free calibration approaches combine post-hoc scaling with conformal guarantees [21].

Complementary to point-estimate calibration, split conformal prediction [28], [29] produces prediction sets with a guaranteed marginal coverage of $1 - \alpha$ without any distributional assumption on the underlying model. The guarantees hold under exchangeability of calibration and test data, which is satisfied in the present study by the split protocol described in Section 4.3.4. Selective classification — the option to abstain from prediction — has been formalized by Geifman and El-Yaniv [30], [31] through risk–coverage curves, which are used in Section 5.10 to test Hypothesis H4.

Chapter 3

Project Context and Scope

3.1 Academic Research Framework

This thesis was conducted within the Final Year Project (*Projet de Fin d'Études*, PFE) of the PGE5 program at aivancity, from March to September 2026. Unlike most PFE projects at aivancity, which are structured around a six-month industry internship in a host company, the present project is a purely academic research contribution. This choice was made in agreement with the academic supervisor, Dr. Anuradha Kar, after a preliminary review of the literature identified a specific methodological gap — the systematic auditing of large language model calibration in clinical trial matching — that could be addressed with academic resources alone, without requiring hospital or industry partners.

The academic framing has three practical consequences for the study. First, the research question and the four pre-registered hypotheses were formulated to be testable with publicly available datasets and open-source models, avoiding any dependency on proprietary systems or restricted patient data. Second, the study operates on a purely retrospective basis: no prospective clinical validation is performed, and no recommendations for real patient enrollment are made. Third, the computational budget is limited to what a single graduate student can access on free-tier cloud platforms, which caps the model sizes at seven to nine billion parameters and shapes several of the design choices described in Chapter 4.

Regular supervision meetings with Dr. Kar were held throughout the project. Progress was formalized through monthly reports covering the scoping phase (March), the state of the art review (April), the experimental pipeline design (May), the execution of the twelve model runs (June), the extended analyses on mode collapse and truly discriminative cases (July), and the writing phase (August–September).

3.2 Origin and Evolution of the Research Question

The initial research question, formulated during the scoping phase, focused on patient-reported uncertainty communication in clinical trial matching: how do LLMs express uncertainty to end users, and does this communication align with what patients expect? A month of preliminary literature review, however, revealed that this framing conflated two distinct research questions — the human factors of uncertainty communication, and the underlying probabilistic reliability of the model outputs — and that only the second could be rigorously addressed within the scope of a Master's thesis.

The question was therefore reformulated to focus exclusively on the probabilistic reliability of LLM outputs. This reformulation offered three advantages. First, it aligned the study with an established research tradition on classifier calibration, providing well-defined metrics (ECE, Brier score, reliability diagrams) and clear evaluation thresholds (Van Calster's 0.05). Second, it made the hypotheses testable with pre-registered statistical procedures, protecting the analysis against confirmation bias. Third, it identified a specific methodological gap in the medical LLM literature: while several papers had evaluated the accuracy of medical LLMs on clinical benchmarks, no systematic audit of their calibration had been performed for the clinical trial matching task. This gap became the specific target of the study.

The final research question — "Are the eligibility scores produced by LLM-based patient–trial matching systems probabilistically reliable, and does RLHF post-training affect this reliability?" — is a narrower, more falsifiable version of the initial intent, and the tradeoff is a deliberate methodological choice.

3.3 Dataset: TrialGPT-Criterion-Annotations

3.3.1 Dataset Description

The empirical foundation of this study is the TrialGPT-Criterion-Annotations dataset, released by Jin et al. [1] alongside their TrialGPT system in *Nature Communications*. The dataset comprises 1,015 patient–criterion pairs, where each pair associates a synthetic patient note with a single eligibility criterion drawn from a real clinical trial listed on ClinicalTrials.gov. Each pair is annotated independently by three physicians (whose consensus label serves as the ground truth) and by GPT-4 (used as an automated second opinion). Both annotation sources apply the same six-level ordinal scale, described in detail in Section 4.3.1.

The dataset is publicly available under a CC BY 4.0 license, hosted on Zenodo alongside the TrialGPT source code. Its release supports the reproducibility of the original TrialGPT evaluation and, by extension, the present audit.

3.3.2 Justification of the Single-Dataset Choice

An important scoping decision of this study is the deliberate reliance on a single benchmark. Three alternatives were considered and rejected during the design phase. First, the TREC Clinical Trials 2021–2022 tracks provide relevance judgments for patient–trial pairs, but they operate at the trial level rather than the criterion level and use only three ordinal grades (definitely relevant, possibly relevant, irrelevant). Adapting the calibration protocol to this coarser granularity would preclude the fine-grained analysis of inclusion versus exclusion criteria that motivates several of our findings. Second, the SIGIR 2016 clinical trial task shares the same limitations as TREC. Third, hospital-derived cohorts (n2c2, MIMIC-derived subsets) would provide realistic patient data but require institutional partnerships and ethics approval that fall outside the academic scope of the project.

TrialGPT-Criterion-Annotations is, to our knowledge, the only publicly available benchmark that combines per-criterion granularity, a six-level ordinal scale, and an explicit inclusion/exclusion typology. These properties are jointly essential for the calibration analysis described in Chapter 4. External validation on other datasets, when feasible, is discussed as future work in Chapter 6.

3.4 Ethical Considerations

The ethical framework of this study is straightforward due to the nature of the dataset. TrialGPT-Criterion-Annotations contains no real patient data: all clinical notes were synthesized by the original authors from publicly available templates on ClinicalTrials.gov, using clinical concept substitution to produce notes that are semantically realistic yet fully non-identifying. No personal health information is processed at any stage of the pipeline, and no institutional review board approval or ethics committee clearance was required.

The absence of real patient data does not eliminate all ethical considerations. The results reported in Chapter 5 concern models that could plausibly be deployed in real clinical screening workflows, and the mode collapse behaviors documented in Section 5.6 would translate into systematic screening failures if left undetected in production. The methodology described in this thesis is therefore intended as a template for pre-deployment audits rather than as a certification of the models themselves for clinical use. Practitioners integrating LLMs into recruitment workflows retain full responsibility for validating the models on their own populations and for maintaining human oversight of eligibility decisions.

Beyond patient data, the study also complies with the licensing terms of the models under evaluation. All six models are released under permissive open-source licenses (Apache 2.0 for Mistral, custom research licenses for Llama-3.1 and Meditron, MIT for BioMistral) that permit non-commercial evaluation and redistribution of inference outputs. The prediction files archived in the project repository (Appendix D) fall within the scope of these licenses.

3.5 Constraints and Resources

3.5.1 Computational Constraints

All experiments were performed on the Kaggle free-tier compute environment, which provides access to a single Tesla T4 GPU with 16 GB of memory and a weekly quota of 30 GPU-hours. This constraint shaped two important design choices. First, the model sizes are capped at seven to nine billion parameters, which excludes larger medical LLMs such as Meditron-70B or Med42-v2-70B from the evaluation. Second, all models are quantized to 4-bit NF4 precision, which reduces the memory footprint from approximately 14 GB per model (in half-precision) to approximately 4 GB. The justification of these choices and their potential impact on the results are discussed in Sections 4.4 and 4.7.

3.5.2 Model Access Constraints

Access to the six selected models is subject to licensing terms that occasionally require prior approval. Llama-3.1 requires acceptance of Meta's community license through the HuggingFace hub, which was obtained during the setup phase. Gemma and Med42-v2, considered but excluded from the final selection, are subject to more restrictive licenses that complicate reproducibility and were partly responsible for their exclusion (Section 4.4.3). All models retained in the final evaluation are freely accessible for research use without additional approval.

3.5.3 Time Constraints

The PFE was conducted over six months, from March to September 2026. The scoping phase (March) established the research question and the literature review. The design phase (April–May) formalized the pre-registered hypotheses and the experimental matrix. The execution phase (June–July) ran the twelve model configurations and the downstream analyses. The writing phase (August–September) produced the present manuscript. The time constraint precludes external validation on additional datasets, prospective clinical validation, and extension to larger model scales; these directions are discussed in Chapter 6.

3.6 Scope of This Work

The scope of this thesis is delimited both by what it includes and by what it deliberately excludes. Within scope, the study contributes: a systematic calibration audit of twelve LLM configurations on a single well-characterized benchmark; a pre-registered test of the effect of RLHF post-training on calibration in the clinical domain; a comparison between generalist and medically specialized open-source models; a documentation of two novel failure modes (mode collapse and complete failure on discriminative cases) that had not been previously reported for these models; and an evaluation of post-hoc recalibration and conformal prediction as practical remedies.

Explicitly out of scope, and deferred to future work in Chapter 6, are: external validation on independent benchmarks such as TREC Clinical Trials; prospective clinical validation involving real patient enrollment; extension to model sizes above nine billion parameters; extension to multi-agent debate architectures as a mitigation for mode collapse; fine-tuning or continued pretraining of the evaluated models; and multilingual evaluation on non-English clinical text.

This delimitation is intended to make the contribution of the thesis falsifiable and specific, rather than exhaustive. The methodology proposed here can be extended in each of the excluded directions, and several such extensions are outlined as concrete research programs in Chapter 6.

Chapter 4

Methodology and Technical Approach

4.1 Overview of the Experimental Design

This chapter describes the methodology used to audit the calibration of large language models in patient–clinical trial matching. The experimental design follows a pre-registered protocol consistent with the TRIPOD+AI reporting guidelines [23], which recommend explicit hypothesis specification, statistical rigor, and reproducibility for clinical prediction models.

The evaluation pipeline is organized around three principles. First, all predictions are generated on a single held-out benchmark, TrialGPT-Criterion-Annotations [1], which contains 1,015 patient–criterion pairs annotated by three physicians and by GPT-4. Second, model probabilities are extracted directly from the log-probabilities of candidate labels rather than from free-form generation, in order to obtain well-defined probability distributions suitable for calibration analysis. Third, every hypothesis test is defined before the analysis begins, and p-values are corrected for multiple comparisons.

The design forms a 2×2 matrix crossing model type and post-training regime. Along one axis, we compare generalist models (Mistral-7B, Llama-3.1-8B) with medically specialized models (Meditron-7B, BioMistral-7B). Along the other axis, we compare base models (without RLHF) with instruction-tuned models (with RLHF). This crossing produces six models, each of which is evaluated with two prompt variants: a generic minimalist prompt (Prompt A) and an adapted prompt with expert guidance (Prompt B). The full factorial yields twelve configurations, all evaluated on the same 1,015 pairs, for a total of 12,180 predictions.

Figure 4.1 summarizes the pipeline. Section 4.2 states the four pre-registered hypotheses. Section 4.3 describes the dataset. Sections 4.4 and 4.5 justify the model and prompt selection. Section 4.6 details the log-probability scoring pipeline. Section 4.8 defines the calibration and validity metrics. Section 4.9 presents the statistical analysis plan, including the recalibration and conformal prediction procedures.

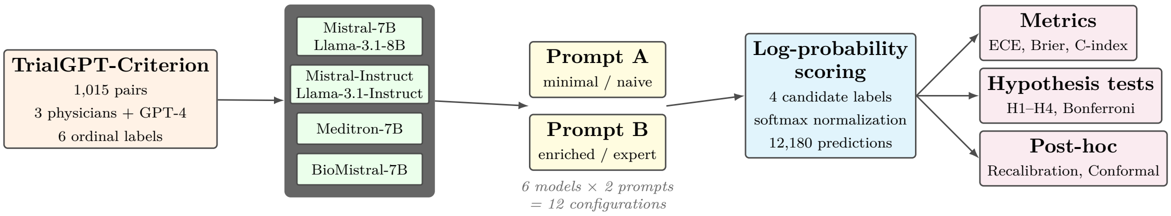

Figure 4.1: Overview of the experimental design. Six open-source LLMs are evaluated with two prompt variants on 1,015 patient–criterion pairs, producing twelve configurations tested against four pre-registered hypotheses.

4.2 Research Question and Pre-registered Hypotheses

The main research question addressed in this work is stated as follows:

> *Are the eligibility scores produced by large language models in patient–clinical trial matching probabilistically reliable, and does reinforcement learning from human feedback affect this reliability?*

This question is decomposed into four falsifiable hypotheses, each associated with a numerical rejection threshold defined prior to the experimental phase. Pre-registration of hypotheses is a standard safeguard against confirmation bias in empirical work [23] and provides a clear criterion for evaluating each finding.

Hypothesis H1 tests whether current LLM pipelines are systematically miscalibrated. The rejection threshold is derived from Van Calster et al. [22], who characterize an Expected Calibration Error (ECE) below 0.05 as evidence of a well-calibrated predictive system in clinical contexts.

Hypothesis 1 (H1: Poor calibration of current LLM pipelines)**.** *The Expected Calibration Error of an LLM-based eligibility scoring pipeline exceeds the well-calibrated threshold of* 0.05*:*

$$ECE(pipeline) > 0.05.$$

Hypothesis H2 extends the finding of Kadavath et al. [32], who observed that RLHF post-training degrades calibration on general question-answering tasks, to the specific domain of clinical trial matching. It is the central original contribution of this thesis. The comparison uses a permutation test on the difference in ECE between paired Base and Instruct versions of the same model family, with Bonferroni correction across families and prompt variants.

Hypothesis 2 (H2: RLHF degrades calibration)**.** *Within a given model family, the*

RLHF-aligned Instruct variant exhibits worse calibration than its Base counterpart:

$$ECE(Instruct) > ECE(Base).$$

Hypothesis H3 tests whether, independently of absolute calibration, the model's scores respect the ordinal structure of the eligibility labels. Ordinal validity is measured by the concordance index (C-index), and the threshold of 0.70 corresponds to conventional discrimination criteria used in clinical prediction modeling [22].

Hypothesis 3 (H3: Ordinal validity)**.** *The C-index between the LLM ordinal score and the expert ordinal label exceeds the standard discrimination threshold:*

$$\text{C-index} > 0.70.$$

Hypothesis H4 addresses the practical value of confidence-based abstention. If confidence scores are meaningful, restricting predictions to the most confident subset should yield higher classification quality. The threshold of +0.02 on macro-F1 gain is chosen to reflect a clinically noticeable improvement without being trivially achievable.

Hypothesis 4 (H4: Selective classification improves quality)**.** *Restricting predictions to the top 80% most confident cases increases macro-F1 by at least* 0.02*:*

$$F_1(coverage = 80\%) > F_1(coverage = 100\%) + 0.02.$$

Together, these four hypotheses probe distinct aspects of probabilistic reliability: absolute calibration (H1), the effect of RLHF (H2), ordinal consistency (H3), and practical usability under abstention (H4). All rejection criteria and statistical tests were fixed prior to running the experiments and are reported without modification in Chapter 5.

4.3 Data Preparation

The experimental evaluation is conducted on the TrialGPT-Criterion-Annotations dataset [1], released alongside the TrialGPT system by Jin et al. To our knowledge, it is the only publicly available benchmark that provides multi-level expert annotations at the level of individual eligibility criteria, with an explicit distinction between inclusion and exclusion criteria. These properties are essential for a fine-grained calibration analysis: aggregating annotations at the trial level, as in TREC Clinical Trials or SIGIR benchmarks, would preclude the per-criterion probability estimation required by the present protocol.

4.3.1 Dataset Composition

The dataset comprises 1,015 patient–criterion pairs. Each pair associates a synthetic patient note with a single eligibility criterion drawn from a real clinical trial. Two independent annotations are provided per pair: a consensus label from three physicians, and a label produced by GPT-4. Both annotations use the same six-level ordinal scale, described in Table 4.1.

Table 4.1: The six eligibility labels used in TrialGPT-Criterion-Annotations, organized by criterion type.

Label	**Criterion type**	**Semantics**
included	inclusion	The patient satisfies the inclusion criterion.
not included	inclusion	The patient does not satisfy the criterion.
excluded	exclusion	The patient has the excluding condition.
not excluded	exclusion	The patient does not have the excluding condition (favorable).
not enough information	both	The note does not permit a decision.
not applicable	both	The criterion does not apply to this patient.

Two of the six labels function as evasion or uncertainty markers rather than decisive clinical judgments ("not enough information" and "not applicable"). We deliberately keep them in the label space to preserve the semantics of the original benchmark, while treating their behavior as an object of study in its own right (see Sections 5.9 and 5.6).

4.3.2 Label Distribution

The distribution of expert annotations is markedly imbalanced (Table 4.2). "Not excluded" dominates at 47.7%, reflecting the fact that most patients do not present the specific medical conditions targeted by exclusion criteria. "Not enough information" accounts for a further 28.9%. The remaining 23.4% of pairs, distributed across four labels, represent the cases requiring a truly discriminative clinical judgment. This asymmetry is critical for the interpretation of results in Chapter 5: aggregate metrics computed over the full 1,015 pairs can be dominated by the two majority classes and may mask complete failure on the discriminative subset.

4.3.3 Textual Characteristics

Patient notes are short and structured: their length ranges from 232 to 1,447 characters with a median of 579 characters. This compactness is important for two reasons. First, it means that no aggressive truncation is required to fit the notes within LLM context

Table 4.2: Expert-label distribution in TrialGPT-Criterion-Annotations.

Label	Count	Proportion
not excluded	484	47.7%
not enough information	293	28.9%
included	74	7.3%
excluded	55	5.4%
not applicable	55	5.4%
not included	54	5.3%
Total	1,015	100.0%

windows, avoiding an additional source of variability. Second, short notes place a heavier reasoning burden on the model: information is often implicit, and the model must infer unstated conditions from the surrounding clinical context. Criterion texts are also short, with a median length of 58 characters, ranging from declarative statements (*Age 18 to 75 years*) to more elaborate clauses involving numeric thresholds or drug interactions.

4.3.4 Split for Recalibration Analysis

The dataset is used in three distinct ways depending on the analysis, each governed by a specific splitting strategy.

For the primary calibration audit (Chapter 5, Section 5.3) and the H2 permutation tests, the models are queried on the full set of 1,015 pairs. No split is required at this stage because the models are used off-the-shelf, without any fitting on the target data. Each of the twelve configurations produces one prediction per pair, and calibration metrics are computed on the full set. The permutation tests reshuffle pair-level assignments between Base and Instruct variants, preserving the identity of individual pairs to maintain paired comparisons.

For the post-hoc recalibration analysis (Section 5.11), we adopt a 70/30 random split. The 70% subset (approximately 710 pairs) is used to fit the recalibration function (isotonic regression, temperature scaling, or Platt scaling), while the 30% subset (approximately 305 pairs) is used to evaluate the recalibrated ECE on data unseen during fitting. The split is stratified by criterion type to preserve the inclusion/exclusion ratio in both subsets and prevent imbalance artifacts. The random seed is fixed at 42 for reproducibility, and the same split is applied consistently across all twelve configurations to enable direct comparison.

To assess whether the recalibration gain observed on a single 70/30 split is stable or an artifact of a favorable split, we complement the primary analysis with a 5-fold cross-validation on the same data. Each fold uses 80% of the pairs for fitting and

20% for evaluation, with the folds rotating through the dataset. We report the mean ECE reduction across the five folds together with its standard deviation, providing a direct measure of the stability of the recalibration effect. This cross-validation is applied to isotonic recalibration only, which is the method retained as the primary recommendation in Chapter 5.

For the analysis of performance on truly discriminative cases (Section 5.7), we introduce a further stratification that separates the 238 pairs whose expert label falls outside the two dominant classes ("not excluded" and "not enough information") from the remaining 777 pairs. This stratification is not a random split but a principled selection driven by the class-imbalance analysis reported in Section 4.3.2, and it is used to reveal model behavior on cases requiring genuine clinical decisions rather than majority-class shortcuts.

4.3.5 Ethical and Reproducibility Considerations

The ethical framework of this study is straightforward due to the nature of the dataset. TrialGPT-Criterion-Annotations contains no real patient data: all clinical notes were synthesized by the original authors [1] to match the structure and semantics of authentic recruitment scenarios without disclosing any personal health information. The synthesis process combined publicly available templates from ClinicalTrials.gov with clinical concept substitution to produce notes that are semantically realistic yet fully non-identifying. No institutional review board approval or ethics committee clearance was therefore required for the present study, since no human subject data is processed.

The dataset is released under a CC BY 4.0 license, permitting redistribution and derivative use for research purposes with attribution. This licensing framework aligns with the reproducibility commitments of this thesis. All prediction files produced during the twelve model runs, comprising the softmax distributions over candidate labels for each of the 1,015 pairs, are preserved as Parquet files and archived in the project repository listed in Appendix D. The complete set of downstream analysis outputs (calibration metrics, permutation test p-values, recalibration results, conformal prediction sets, and mode collapse statistics) is also archived as CSV files to enable independent verification of every numerical result reported in Chapter 5.

Beyond data ethics, the study raises considerations related to the deployment of the audited models in clinical practice. The results reported in Chapter 5 are informative but not directly actionable: no model tested reaches a level of probabilistic reliability sufficient for autonomous eligibility screening, and the mode collapse behaviors documented in Section 5.6 would in a clinical setting translate into systematic screening failures if left undetected. These findings underscore the responsibility of practitioners who consider integrating LLMs into recruitment workflows to conduct calibration audits before deployment and to interpret model outputs jointly with human clinical judgment rather than as replacements for it. The methodology described in this chapter is intended as a practical template for such pre-deployment audits.

4.4 Model Selection

The choice of models under evaluation is the central design decision of this work. Rather than surveying a broad catalogue of open-source LLMs, we prioritize a small selection organized as a controlled experimental matrix. This design enables clean isolation of the effects under study — particularly the effect of RLHF post-training — while remaining feasible on the limited computational resources available.

4.4.1 Selection Criteria

Three criteria guide the selection of models. First, the model must be available in both a base and an instruction-tuned variant with a common pretraining, since testing hypothesis H2 requires paired comparisons within the same model family. Second, the selection must span the two-by-two matrix of interest: generalist versus medically specialized pretraining, crossed with the presence or absence of RLHF post-training. Third, the model must fit into 16 GB of GPU memory when quantized to 4 bits, which is the memory budget of a free-tier T4 GPU on Kaggle. This constraint effectively caps the parameter count at 7–9 billion.

These three criteria eliminate several candidates that would otherwise be attractive. Models larger than 9 B parameters (Meditron-70B, Med42-v2-70B, Llama-3.3-70B) exceed the memory budget. Model families without a clean base–instruct pairing (Med42-v2 releases only the instruction-tuned variant) preclude the H2 comparison. Model families released without a medical counterpart (Gemma, DeepSeek) do not add information for our two-by-two matrix.

4.4.2 The Six Selected Models

The six selected models, together with their scientific role in the experimental matrix, are summarized in Table 4.3.

Mistral-7B-v0.3 and its instruction-tuned counterpart Mistral-7B-Instruct-v0.3 form the primary base–instruct pair for testing H2. The two models share the same pretraining corpus and architecture; the Instruct variant differs only through subsequent supervised fine-tuning and RLHF. This clean isolation of the RLHF stage makes them the reference comparison for our central hypothesis.

Table 4.3: Selected models and their scientific role in the two-by-two experimental matrix.

Model	Type	Scientific Role
Mistral-7B-v0.3	Generalist, no RLHF	H2 control
Mistral-7B-Instruct-v0.3	Generalist, with RLHF	H2 test
Llama-3.1-8B	Generalist, no RLHF	H2 replication
Llama-3.1-8B-Instruct	Generalist, with RLHF	H2 replication
Meditron-7B	Medical, no RLHF	Pure specialization effect
BioMistral-7B	Medical, inherited RLHF	Validation vs. Labrak et al. (2024)

Llama-3.1-8B and Llama-3.1-8B-Instruct play the same role for a second, independent model family. Testing H2 on two families rather than one rules out the possibility that any observed effect is specific to a single training pipeline. Llama-3.1 also brings a different tokenizer, a larger context window, and a different alignment methodology, which strengthens the generality of the conclusions.

Meditron-7B [33] is a foundation model pretrained on 46,000 clinical practice guidelines and a large corpus of PubMed abstracts, starting from Llama-2 weights. Crucially, no official instruction-tuned variant of Meditron-7B has been released; the base model is the only publicly available version. Meditron therefore isolates the effect of medical pretraining without any confound with RLHF, providing a direct test of whether medical specialization improves calibration in the absence of instruction alignment.

BioMistral-7B [34] occupies the opposite corner of the two-by-two matrix. It is derived from Mistral-7B-Instruct-v0.1 through continued pretraining on PubMed Central Open Access texts. It therefore inherits the RLHF alignment of its Mistral-Instruct parent while adding a medical specialization stage. BioMistral is also the only open-source medical LLM with published ECE values, which provides an external validation point for our measurement methodology.

Together, these six models populate the two-by-two matrix completely (Table 4.4). Each of the three scientific questions associated with the matrix — the effect of RLHF, the effect of medical specialization, and their interaction — can be tested through paired comparisons within this design.

Table 4.4: The two-by-two experimental matrix of selected models.

	Without RLHF	With RLHF
Generalist	Mistral-7B Llama-3.1-8B	Mistral-7B-Instruct Llama-3.1-8B-Instruct
Medical	Meditron-7B	BioMistral-7B

4.4.3 Excluded Models

Several candidate models were considered and deliberately excluded from the final selection. Qwen2.5-7B and DeepSeek-V2-Lite were excluded as redundant additions to the generalist axis: they would not provide new evidence for H2 beyond what Mistral and Llama already establish. Gemma-2-9B was excluded for the same reason, and additionally because its license restricts commercial redistribution of derivative model outputs, complicating reproducibility.

Among medical LLMs, Med42-v2 [35] and OpenBioLLM-8B were excluded because their base versions are not publicly released, precluding a within-family H2 comparison. HuatuoGPT-o1 [36] was excluded because its training corpus is primarily Chinese, which introduces a confound with the English-only nature of the TrialGPT-Criterion benchmark. Larger medical models (Meditron-70B, Med42-v2-70B, Aloe-70B) all exceed the memory budget of a free-tier T4 GPU, even at 4-bit quantization.

The rationale for this restrictive selection is that a small number of well-matched models tested on the same protocol yields more interpretable conclusions than a broader benchmark that would necessarily mix confounding factors. Extending the analysis to additional families is a natural direction for future work, discussed in Chapter 6.

4.5 Prompt Design

The prompt used to elicit predictions from a large language model is not a neutral component of an evaluation pipeline: it is a design choice that directly affects the model's outputs. Any calibration audit that fixes a single prompt inevitably conflates two effects — the properties of the model and the properties of the prompt — which cannot be separated post hoc. To address this ambiguity, we evaluate each of the six models under two distinct prompt variants, treating the prompt as an experimental variable rather than a fixed convention.

4.5.1 Rationale for Two Prompt Variants

The two prompts are designed to represent the two extremes of a spectrum of elicitation strategies commonly encountered in practice. The first (Prompt A) is intentionally minimal: it presents the classification task and the label vocabulary without any explanation, in the manner of a naive user querying the model. The second (Prompt B) is enriched with definitions, examples, and explicit reasoning rules, in the manner of an expert operator who invests time in prompt engineering. By evaluating each model with both prompts, we obtain four independent measurement points per model family for hypothesis H2 (Base–Instruct comparison at Prompt A, Base–Instruct comparison at Prompt B), doubling the statistical power of the RLHF test.

This design also enables three additional analyses beyond the pre-registered hypotheses. First, the effect of prompt design itself becomes measurable within each model. Second, the interaction between RLHF and prompt design becomes testable: some models may be more sensitive to prompt enrichment than others. Third, the design places our results in a stronger position with respect to external validity: a finding replicated under two different prompts is more robust than a finding measured under a single one.

4.5.2 Prompt A: Generic Baseline

Prompt A is a minimalist prompt containing only the classification instruction, the input note, the criterion text, and the list of candidate labels. It does not define the labels, does not provide examples, and does not include reasoning rules. Its purpose is to establish a baseline that corresponds to how a first-time user would query the model, and to provide a reference point against which enriched prompts can be compared.

The complete text of Prompt A is given in Appendix B. Its essential structure is:

> *Classify whether this patient meets the eligibility criterion.*
> *Patient: [note]*
> *Criterion ([type]): [criterion]*
> *Choose exactly one label from: included, not included, excluded, not excluded, not enough information, not applicable.*
> *Answer:*

Prompt A is applied identically to all six models. For instruction-tuned models, the prompt is wrapped in the model's native chat template. For base models, the prompt is presented as a plain-text continuation task without special tokens.

4.5.3 Prompt B: Adapted with Expert Guidance

Prompt B is an adapted prompt containing definitions of the six labels, one worked example per label, and a critical inference rule for exclusion criteria that addresses a subtle failure mode observed in preliminary experiments. The full text is reproduced in Appendix B.

Three design choices distinguish Prompt B from Prompt A. First, each of the six labels is defined with an explicit semantic description. For example, "not included" is characterized as "The patient note contains information showing the patient does NOT meet this inclusion criterion," which distinguishes it from "not enough information" (where the note is silent). Second, a one-sentence example is provided for each label, drawn from realistic clinical situations. Third, a specific inference rule is stated for exclusion criteria:

> *If the criterion names a specific disease, condition, or medical history, and the note does not mention it, the default is "not excluded" (the patient most likely does not have it). Reserve "not enough information" only for criteria requiring a specific lab value or measurement that is truly missing.*

This rule directly targets the tendency of language models to produce "not enough information" for any exclusion criterion whose target condition is not explicitly discussed in the patient note — a pattern that mechanically converts the absence of a mention into an expression of epistemic uncertainty, when clinically it should be treated as evidence of absence.

4.5.4 Adaptations for Base vs. Instruct Models

Base models and instruction-tuned models expect prompts in different formats, and applying the same prompt verbatim to both would create an artificial handicap for the base variant. To ensure a fair comparison, we adapt the surface form of each prompt to the target model class while preserving its semantic content.

For instruction-tuned models, prompts are formatted using the model's native chat template, which delimits user and assistant roles with model-specific special tokens (`[INST]` for Mistral, `<|start_header_id|>` for Llama-3). This is the format for which these models were aligned and where they behave as expected.

For base models, no chat template is applied. Instead, the prompt includes one or more few-shot examples that establish the desired input-output pattern by demonstration. Prompt A for base models includes a single example; Prompt B includes three examples covering distinct label types. This asymmetry reflects a genuine limitation of base models — they do not natively follow instructions — and treating it as an evaluation artifact would misrepresent how these models are actually used in practice. The introduction of few-shot examples is nevertheless a potential confounder for cross-family comparisons, and we discuss it explicitly in Chapter 6.

The complete text of all four prompt variants (Prompt A / Prompt B, Instruct / Base) is provided in Appendix B to support independent reproduction of the pipeline.

4.6 Scoring Pipeline: Log-Probability Extraction

Standard practice for querying a large language model on a classification task is to sample a textual output and post-process it to recover a label. This approach has two limitations for a calibration audit. First, the sampled output is a point estimate: it discards the underlying probability distribution over candidate labels, which is precisely the object of interest for calibration analysis. Second, generation is stochastic, introducing an additional source of variability that would complicate the reproducibility of downstream statistics. We therefore adopt a log-probability scoring approach that directly reads out the model's probability distribution over the candidate labels, without any generation step.

4.6.1 Principle of Log-Probability Scoring

For a given patient–criterion pair, we first construct the prompt as described in Section 4.5. Rather than sampling a continuation, we consider each candidate label as a hypothetical completion of the prompt. For each label ℓ, we compute the mean log-probability of its token sequence conditional on the prompt:

$$s_\ell = \frac{1}{|\ell|} \sum_{t=1}^{|\ell|} \log p(\ell_t \mid \text{prompt}, \ell_{<t}) \tag{4.1}$$

where ℓ_t is the t-th token of the label and $|\ell|$ is the total number of tokens. Length normalization by $|\ell|$ is necessary because labels have unequal token lengths in the tokenizer: without it, longer labels such as "not enough information" would be systematically penalized against shorter ones such as "included".

The resulting scores $\{s_\ell\}$ are then normalized through a softmax to obtain a proper probability distribution π over the candidate labels:

$$\pi_\ell = \frac{\exp(s_\ell)}{\sum_{\ell' \in \mathcal{L}} \exp(s_{\ell'})} \tag{4.2}$$

The predicted label $\hat{y}$ is the argmax of π, and the associated confidence $\hat{p}$ is the maximum probability. Crucially, the full distribution π is retained for downstream analyses, including recalibration, conformal prediction, and the simulation of constrained label spaces (Section 5.6).

4.6.2 Restriction of the Candidate Set by Criterion Type

The six labels described in Section 4.3 split into two groups by criterion type. For inclusion criteria, the semantically valid labels are *included*, *not included*, *not enough* *information*, and *not applicable*. For exclusion criteria, they are *excluded*, *not excluded*, *not enough information*, and *not applicable*. In both cases, four labels are candidates and two would be semantically nonsensical (e.g., predicting "included" for an exclusion criterion). We therefore restrict the softmax in Equation 4.2 to the four candidates appropriate to the criterion type at hand. This restriction preserves the ordinal structure of the task and prevents the model from being penalized for probability mass placed on labels that could not have been correct in the first place.

4.6.3 Algorithmic Summary

Algorithm 1 summarizes the full scoring procedure applied to each patient–criterion pair.

Algorithm 1 Log-probability scoring for eligibility label prediction

Require: Patient note n, criterion c, criterion type t
Ensure: Predicted label $\hat{y}$, confidence $\hat{p}$, distribution π

```
Build prompt from n, c, t
L ← candidate labels according to t
for all ℓ ∈ L do
    s_ℓ ← mean log-probability of tokens of ℓ
end for
π ← softmax(s)
ŷ ← arg max_ℓ π_ℓ
p̂ ← π_ŷ
return ŷ, p̂, π
```

Because no sampling is involved, the procedure is fully deterministic: running the same model on the same input twice yields identical scores. This determinism is important for the reproducibility of the entire downstream analysis, including the bootstrap confidence intervals and permutation tests described in Section 4.9.

4.6.4 Implementation Notes

The scoring loop is implemented using the standard causal language modeling interface of the HuggingFace `transformers` library. For a given prompt, we tokenize the concatenation of the prompt and each candidate label separately, run a single forward pass per concatenation, and read out the token log-probabilities from the model's logits after applying a log-softmax over the vocabulary. This approach requires four forward passes per patient–criterion pair (one per candidate label), which is approximately twice as expensive as free-form generation of a short answer. In practice, on a T4 GPU with 4-bit quantization, the four passes complete in approximately three seconds per pair for a 7-billion-parameter model, and the full evaluation of 1,015 pairs takes fifty to seventy minutes depending on the model.

4.7 Quantization and Infrastructure

Running 7-to-9-billion-parameter models on a free-tier T4 GPU (16 GB of memory) requires reducing the memory footprint of the model weights. We apply 4-bit NF4 quantization [37] through the `bitsandbytes` library, which represents each weight as a 4-bit value drawn from a distribution matched to the empirical distribution of neural network weights. This reduces the memory required to store a 7-billion-parameter model from approximately 14 GB (in half-precision) to approximately 4 GB, leaving sufficient headroom for the key-value cache during inference.

Prior work has shown that 4-bit NF4 quantization introduces negligible degradation on downstream evaluation benchmarks compared to full precision [37], including on tasks that involve long-form generation or fine-grained probability estimation. We rely on this evidence to justify the use of NF4 quantization for our calibration audit, while acknowledging that a small residual effect of quantization on absolute ECE values cannot be entirely ruled out. Since Hypothesis H2 and the associated Bonferroni-corrected permutation tests compare *differences* in ECE between paired variants evaluated under the same quantization regime, any systematic bias introduced by quantization cancels out and does not affect the conclusions.

All experiments were run on Kaggle notebooks with a single Tesla T4 GPU. The compute quota (30 GPU-hours per week on the free tier) was sufficient to complete the twelve runs and all downstream analyses within the experimental phase of the project.

4.8 Evaluation Metrics

The evaluation combines calibration metrics, classification metrics, ordinal validity metrics, and behavioral metrics. Each captures a distinct aspect of probabilistic reliability, and no single metric is sufficient on its own. This section defines every metric used in Chapter 5.

4.8.1 Expected Calibration Error (ECE)

The Expected Calibration Error, introduced by Naeini et al. and popularized by Guo et al. [17], measures the average absolute gap between predicted confidence and observed accuracy across confidence bins:

$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{n} \left|\mathrm{acc}(B_k) - \mathrm{conf}(B_k)\right| \tag{4.3}$$

where predictions are partitioned into K equal-width bins $B_1, \ldots, B_K$ based on their confidence, n is the total number of predictions, $\mathrm{acc}(B_k)$ is the empirical accuracy of predictions in bin B_k, and $\mathrm{conf}(B_k)$ is the average confidence of predictions in the same bin. A perfectly calibrated model satisfies $\mathrm{acc}(B_k) = \mathrm{conf}(B_k)$ for all bins, yielding an ECE of zero. We use $K = 10$ equal-width bins on $[0, 1]$.

4.8.2 Brier Score

The Brier score [20] measures the mean squared error between predicted probability and the true binary outcome (correct or incorrect):

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2 \tag{4.4}$$

where p_i is the predicted confidence and $y_i \in \{0, 1\}$ indicates whether the prediction was correct. Unlike ECE, which measures calibration alone, the Brier score jointly reflects calibration and sharpness: a model that is both accurate and confident achieves a lower Brier score than one that is only well calibrated but uninformative.

4.8.3 Classification Metrics

We report accuracy as the proportion of predictions matching the expert label, and macro-F1 as the unweighted mean of per-class F1 scores. Macro-F1 is critical in the presence of the severe class imbalance documented in Section 4.3.2: a model that always predicts the majority class achieves an accuracy of 47.7% but a macro-F1 close to zero on the minority classes. This distinction becomes central in Chapter 5 when comparing model behavior across cases of varying clinical difficulty.

4.8.4 C-index for Ordinal Validity

Because the eligibility labels form an ordinal scale, we complement calibration and classification metrics with the concordance index (C-index), which measures the probability that, for a randomly chosen pair of cases with different expert labels, the model correctly ranks them:

$$\text{C-index} = \frac{|\{(i,j) : y_i < y_j \text{ and } s_i < s_j\}|}{|\{(i,j) : y_i \neq y_j\}|} \tag{4.5}$$

where y_i, y_j are ordinal expert labels of cases i and j, and s_i, s_j are the model's scalar scores obtained by mapping the predicted label and confidence onto the ordinal scale. A C-index of 1 indicates perfect ranking, 0.5 indicates random ranking, and the conventional threshold of 0.70 is used in clinical prediction modeling as evidence of acceptable discrimination [22].

4.8.5 Weighted Kappa for Expert-Model Agreement

To measure the agreement between physician annotations and GPT-4 annotations in the pre-LLM audit (Section 5.2), we use Cohen's weighted kappa with quadratic weights [38]. Weighted kappa extends the standard kappa coefficient by penalizing disagreements proportionally to their ordinal distance: a confusion between adjacent labels (e.g., "included" and "not enough information") counts less than a confusion between distant labels (e.g., "included" and "excluded"). We report values in the range $[-1, 1]$, with 1 indicating perfect agreement, 0 indicating chance agreement, and negative values indicating systematic disagreement.

4.8.6 Overconfidence and Prediction Diversity

Beyond calibration and classification, two behavioral metrics quantify how a model uses its probability distribution.

The mean overconfidence is defined as the difference between the average confidence and the average accuracy:

$$\text{overconfidence} = \bar{p} - \text{acc} \tag{4.6}$$

where $\bar{p}$ is the mean predicted confidence across the evaluation set. A positive value indicates that the model tends to overstate its certainty, a negative value that it understates it. Unlike ECE, which is a magnitude, overconfidence is a signed quantity that reveals the direction of the miscalibration.

Prediction diversity is measured as the normalized Shannon entropy of the empirical label distribution:

$$\text{diversity} = \frac{-\sum_{\ell \in \mathcal{L}} f_\ell \log_2 f_\ell}{\log_2 |\mathcal{L}|} \tag{4.7}$$

where f_ℓ is the empirical frequency of label ℓ in the model's predictions and $\mathcal{L}$ is the label vocabulary. A diversity of 0 indicates that the model predicts a single label; a diversity of 1 indicates a uniform distribution over all labels. This metric is central to detecting the mode collapse behavior documented in Section 5.6, which the calibration and classification metrics can mask.

4.8.7 Composite Utility Score

To provide a single rank-ordering of the twelve configurations that reflects clinical usability, we propose a composite utility score combining calibration quality, discrimination, and prediction diversity:

$$\text{utility} = F_1^{\text{macro}} \times (1 - \text{ECE}) \times \text{diversity} \tag{4.8}$$

The three factors are multiplicative rather than additive so that a failure on any single dimension zeroes out the score. A model with excellent calibration but complete mode collapse (diversity = 0) receives a utility score of zero, correctly capturing its lack of practical value. This composite score complements rather than replaces the individual metrics, which remain the primary object of analysis.

4.8.8 Reliability Diagrams

Reliability diagrams complement ECE by visualizing the calibration curve directly. For each of the K bins used in the ECE computation, we plot the average confidence on the horizontal axis and the observed accuracy on the vertical axis. A perfectly calibrated model produces points that lie on the diagonal $y = x$. Points below the diagonal indicate overconfidence; points above indicate underconfidence.

4.8.9 Bootstrap Confidence Intervals

For every point estimate of ECE, Brier score, and C-index, we report a 95% confidence interval obtained by 1,000 bootstrap resamples of the prediction set. Bootstrap confidence intervals do not assume any specific distributional form and are appropriate for the non-Gaussian statistics under study. They also provide a natural way to assess whether a given model's ECE genuinely exceeds the 0.05 threshold of Hypothesis H1, or whether the excess is within sampling noise.

4.9 Statistical Analysis Plan

The four pre-registered hypotheses described in Section 4.2 are tested according to a fixed analysis plan. This section describes the tests, the correction for multiple comparisons, and the additional analyses (post-hoc recalibration and conformal prediction) that complement the primary hypotheses.

4.9.1 Pre-registration of Hypotheses

All four rejection criteria (H1: ECE > 0.05; H2: ECE(Instruct) $>$ ECE(Base); H3: C-index > 0.70; H4: $F_1(80\%) > F_1(100\%){+}0.02$) were fixed before the twelve model runs were executed. This pre-registration protects the analysis against post-hoc adjustment of the criteria to fit the observed results and follows the recommendations of the TRIPOD+AI reporting guidelines [23]. Results are reported without modification of the rejection criteria, and any findings not covered by these four hypotheses are reported separately as original observations (see Chapter 5).

4.9.2 Permutation Tests

Hypothesis H2 compares the ECE of paired Base and Instruct variants of the same model family under the same prompt. Rather than assume any particular distributional form for the ECE difference, we use a permutation test: for 1,000 random permutations of the model labels (Base vs. Instruct) within each pair, we recompute the ECE difference and count the fraction of permutations in which the observed difference is equalled or exceeded. This fraction is the one-sided p-value for the null hypothesis that the two variants have the same underlying calibration. The one-sided formulation is appropriate because Hypothesis H2 is directional: we test whether Instruct is worse than Base, not merely different from it.

4.9.3 Bonferroni Correction

Hypothesis H2 is tested four times (Mistral–Prompt A, Mistral–Prompt B, Llama–Prompt A, Llama–Prompt B) and further comparisons for medical specialization contribute additional tests. To control the family-wise error rate across the eight planned comparisons, we apply a Bonferroni correction with $\alpha_{\text{corrected}} = 0.05/8 = 0.00625$. This correction is conservative but appropriate given the small number of comparisons and the desire for a stringent test of the central hypothesis.

4.9.4 Post-hoc Recalibration Methods

Beyond the primary calibration audit, we evaluate three post-hoc recalibration methods that could correct miscalibration without retraining the model. Temperature scaling [17] applies a single learned scalar T to the logits before the softmax, effectively softening (or sharpening) the confidence distribution. Isotonic regression fits a monotone piecewise-constant function mapping uncorrected confidences to corrected ones; being non-parametric, it can correct arbitrary shapes of miscalibration but requires more data than temperature scaling. Platt scaling fits a logistic function of the uncorrected confidence, offering a compromise between the flexibility of isotonic regression and the parsimony of temperature scaling.

Each method is fitted on the 70% training subset and evaluated on the 30% held-out subset (see Section 4.3.4). To assess whether the recalibration gains are stable across different data splits, we additionally perform a 5-fold cross-validation, reporting the mean and standard deviation of the ECE reduction over the five folds.

4.9.5 Conformal Prediction

To complement point-estimate calibration, we apply split conformal prediction [28], a distribution-free framework that produces prediction sets with a guaranteed marginal coverage rate. Given a target coverage level $1 - \alpha$ (we use $\alpha = 0.10$, i.e., 90% coverage), the framework calibrates a threshold on the non-conformity scores of a held-out calibration set and returns, for each test case, the set of labels whose non-conformity is below that threshold. The size of the prediction set is an informative quantity: large sets indicate high aleatoric uncertainty, while small sets indicate high model confidence. We report both empirical coverage (which should match the target 90%) and average set size.

4.10 Reproducibility

Reproducibility is treated as a first-class methodological requirement throughout this study, following the recommendations of the TRIPOD+AI guidelines [23] and the broader movement toward computational transparency in machine learning research. This section documents the specific measures that support independent replication of the results reported in Chapter 5.

4.10.1 Code Availability

All code produced during this study is archived in a public repository listed in Appendix D. The repository is organized into three components. First, a single Kaggle notebook contains the twelve model runs, from data loading through prediction generation, with inline documentation of every design choice. Second, a set of Python scripts performs the downstream analyses (calibration audit, hypothesis tests, recalibration, conformal prediction, mode collapse analysis, and composite utility scoring) from the archived prediction files. Third, the notebook and scripts share a single `requirements.txt` file that pins the exact versions of every dependency, together with the CUDA version required for GPU acceleration.

4.10.2 Deterministic Pipeline

The full pipeline is designed to be deterministic. The 4-bit NF4 quantization applied to the model weights is itself deterministic: loading the same model with the same quantization configuration produces identical weight matrices across runs. The log-probability scoring procedure described in Section 4.6 involves no sampling and reads out probabilities directly from the model's forward pass, which is also deterministic given fixed weights and inputs. All random operations used in the downstream analyses — the bootstrap resamples for confidence intervals, the permutation tests for Hypothesis H2, the 70/30 splits for recalibration, and the 5-fold cross-validation — use a fixed random seed (42), ensuring that statistical results are bit-exact reproducible across independent runs of the analysis scripts.

4.10.3 Data and Prediction Archives

Two categories of data files are preserved in the project repository. The first category comprises the raw prediction files: for each of the twelve configurations, a Parquet file records the softmax distribution over candidate labels, the predicted label, the associated confidence, the expert label, the criterion type, and metadata linking each prediction to the original patient–criterion pair. These twelve files allow any downstream analysis to be reproduced without re-running the model inference, which is the most computationally expensive step. The second category comprises the analysis outputs: CSV files containing the calibration metrics, permutation test results, recalibration comparisons, conformal prediction sets, mode collapse statistics, and composite utility scores. Each numerical result reported in Chapter 5 can be traced to a specific row in one of these files.

4.10.4 Prompt Verbatim

Because prompt engineering constitutes an experimental variable in this study (Section 4.5), the exact text of all four prompt variants — Prompt A and Prompt B, in both Instruct and Base formulations — is reproduced verbatim in Appendix B. This includes the chat template markers used for instruction-tuned models and the few-shot examples embedded in the base-model prompts. Any deviation from these verbatim prompts constitutes a distinct experimental condition and cannot be considered a replication of the present results.

4.10.5 Computational Requirements

Independent reproduction of the full pipeline requires access to a GPU with at least 16 GB of memory (a Tesla T4 or equivalent) and approximately ten to fifteen hours of

GPU time to complete the twelve model runs. The downstream analyses run on CPU in a few minutes on standard hardware. All experiments reported in this thesis were performed on the Kaggle free-tier compute environment, which is publicly accessible and imposes no barriers to independent replication. The complete list of hardware and software versions, together with the commit hashes of the notebook and analysis scripts at the time of submission, is provided in Appendix D.

4.10.6 Reporting Standards

The reporting in Chapter 5 follows the structure recommended by TRIPOD+AI for prediction model studies: hypotheses are stated before results, the four pre-registered hypotheses are reported alongside their pre-specified rejection thresholds, statistical tests are reported with their exact p-values and their Bonferroni-corrected significance thresholds, and all point estimates are accompanied by 95% bootstrap confidence intervals. Findings not covered by the pre-registered hypotheses are clearly labelled as original observations to distinguish confirmatory from exploratory results. This structured reporting is intended to make the strength of the evidence transparent to readers without requiring them to re-analyze the underlying data.

Chapter 5

Results and Analysis

5.1 Chapter Overview

This chapter reports the empirical findings of the study. The presentation is organized to distinguish confirmatory results (tests of the four pre-registered hypotheses stated in Section 4.2) from exploratory findings (novel observations that emerged during the analysis and are labelled here as original findings F1 through F9).

Sections 5.2 through 5.4 present the pre-LLM audit and the two pre-registered hypotheses that receive the strongest empirical support: H1 (systematic miscalibration, supported across all twelve configurations) and H2 (RLHF degrades calibration, supported by four independent permutation tests after Bonferroni correction). H2 is the central pre-registered contribution of this work.

Sections 5.5 through 5.7 report the three most novel findings, none of which was anticipated in the pre-registration. The enriched-prompt paradox (Section 5.5) shows that expert prompt engineering degrades calibration in five of six models. Mode collapse (Section 5.6) documents that five of the twelve configurations concentrate more than 97% of their predictions on a single label, invalidating the standard interpretation of their apparently reasonable ECE values. Performance on truly discriminative cases (Section 5.7) reveals that both medical LLMs tested achieve 0% accuracy on the 238 cases requiring a genuine clinical decision, despite their apparent superiority on global accuracy.

Sections 5.8 through 5.10 report secondary findings on the inclusion/exclusion asymmetry, model-specific evasion strategies, and the two pre-registered hypotheses that receive partial or no support (H3 on ordinal validity, H4 on selective classification). Sections 5.11 through 5.13 present the post-hoc solutions — isotonic recalibration and conformal prediction — and the composite utility score that provides a single rank-ordering of the twelve configurations. Section 5.14 synthesizes the outcomes of the four hypotheses and the nine original findings.

Every point estimate reported in this chapter is accompanied by a 95% bootstrap confidence interval, and every hypothesis test is reported with its Bonferroni-corrected significance threshold. Detailed tables, per-configuration reliability diagrams, and the full mode collapse analysis are provided in Appendices A, C, and F.

5.2 Pre-LLM Audit and Expert Agreement

Before evaluating the twelve LLM configurations, we conduct a pre-LLM audit of the TrialGPT-Criterion-Annotations dataset itself. This audit serves two purposes: to establish the quality of the ground truth against which the LLMs will be compared, and to expose a systematic annotation bias in GPT-4 that has not been previously documented.

5.2.1 Expert-GPT-4 Agreement

The dataset provides two independent annotations per pair: a consensus label from three physicians (which we treat as the ground truth throughout this thesis) and a label produced by GPT-4. The weighted kappa [38] between the two annotations, computed with quadratic weights on the ordinal scale, is $\kappa_w = 0.819$ (95% bootstrap CI: [0.774, 0.86]). By conventional criteria for inter-rater agreement, this value falls in the range of substantial to almost perfect agreement, confirming that GPT-4 reproduces the physician consensus with high fidelity at the aggregate level.

This aggregate agreement, however, masks a systematic bias visible only at the label level. Table 5.1 reports, for each of the six labels, the ratio of GPT-4 usage rate to expert usage rate. A ratio of 1.0 indicates that GPT-4 and experts use the label with the same frequency; a ratio above 1.0 indicates over-use by GPT-4; a ratio below 1.0 indicates under-use.

Table 5.1: Usage-rate ratios of the six labels (GPT-4 vs. physician consensus). A ratio of 1.0 indicates balanced usage; ratios far from 1.0 reveal systematic annotation bias.

Label	Expert freq.	GPT-4 freq.	Ratio
not applicable	5.4%	11.7%	2.16
not enough information	28.9%	28.6%	0.99
included	7.3%	6.9%	0.95
not included	5.3%	5.0%	0.94
excluded	5.4%	5.5%	1.02
not excluded	47.7%	42.3%	0.89

Five of the six labels are used by GPT-4 within 10% of the physician frequency, in line with the substantial overall kappa. The exception is “not applicable”, which GPT-4 uses at 2.16 times the physician rate. This asymmetric bias — systematic over-use of one specific evasion label rather than a general tendency to over-use uncertainty — is not reported in the original TrialGPT paper [1], where GPT-4 is used as an annotation tool without such an audit. The bias is substantively meaningful: "not applicable" functions as a legitimate escape route when a criterion genuinely does not apply to a patient's situation (e.g., a pregnancy exclusion criterion applied to a male patient), but its over-use by GPT-4 suggests that the model treats a broader class of ambiguous cases as "not applicable" when a decisive label would have been warranted.

Finding 1 (F1: GPT-4's Selective Evasion Bias)**.** *GPT-4 over-uses the "not applicable" label at 2.16 times the physician rate, while using "not enough information" at the same rate as the experts (ratio 0.99). This asymmetric pattern reveals a selective evasion strategy that is not detected by aggregate agreement metrics.*

This finding is meaningful for the downstream analysis of open-source LLMs in Section 5.9, where we show that open-source models exhibit a different but equally systematic evasion pattern — over-use of "not enough information" rather than "not applicable". The two evasion channels appear to be model-specific, likely reflecting distinct alignment strategies during post-training.

5.2.2 Ground Truth Choice

Given the systematic bias identified in F1, we use the physician consensus rather than GPT-4 annotations as the ground truth throughout this thesis. This choice has three consequences. First, GPT-4 is not a model under evaluation in the calibration audit itself: its labels serve only as the input for F1. Second, the accuracy figures reported in subsequent sections are computed against expert labels, which increases their clinical relevance. Third, the class imbalance documented in Section 4.3.2 refers to the expert distribution and is preserved unchanged for the calibration analysis.

5.3 Calibration Audit: H1 Supported Across All Configurations

Hypothesis H1 predicts that the Expected Calibration Error of an LLM-based eligibility scoring pipeline exceeds the well-calibrated threshold of 0.05 [22]. We test this hypothesis across all twelve configurations of the two-by-two matrix defined in Section 4.4, using the metrics and bootstrap protocol described in Section 4.8.

5.3.1 Overall Calibration Results

Table 5.2 reports the ECE, its 95% bootstrap confidence interval, and the Brier score for each of the twelve configurations. All twelve values fall well above the calibration threshold, with lower bounds of the 95% intervals also strictly greater than 0.05. Hypothesis H1 is therefore supported at the level of every individual configuration, not merely in aggregate.

Table 5.2: Calibration metrics across the twelve configurations. All ECE values exceed the 0.05 threshold, with the lower bound of the 95% bootstrap confidence interval also strictly above 0.05 in every case. Configurations are grouped by model family.

Configuration	**ECE**	**95% CI**	**Brier**
Mistral-7B family			
Mistral-Base A	0.133	[0.104, 0.163]	0.258
Mistral-Base B	0.189	[0.161, 0.218]	0.269
Mistral-Instruct A	0.322	[0.297, 0.350]	0.311
Mistral-Instruct B	0.253	[0.229, 0.287]	0.353
Llama-3.1-8B family			
Llama-Base A	0.124	[0.100, 0.148]	0.187
Llama-Base B	0.151	[0.123, 0.177]	0.225
Llama-Instruct A	0.510	[0.483, 0.535]	0.456
Llama-Instruct B	0.602	[0.575, 0.628]	0.558
Medical LLMs			
Meditron A	0.182	[0.153, 0.212]	0.263
Meditron B	0.231	[0.204, 0.259]	0.283
BioMistral A	0.155	[0.126, 0.182]	0.234
BioMistral B	0.194	[0.166, 0.222]	0.244

The ECE values span a wide range, from 0.124 (Llama-Base A) to 0.602 (Llama-Instruct B). Even the best-calibrated configuration is more than twice the clinical threshold, and the worst is more than twelve times above it. The Brier score follows a similar ordering, confirming that the miscalibration is not compensated by favorable sharpness properties.

H1: Systematic Miscalibration Confirmed. All twelve configurations exhibit an ECE strictly greater than 0.05, with the lower bounds of their 95% bootstrap confidence intervals also strictly above the threshold. Hypothesis H1 is supported across the entire experimental matrix.

5.3.2 Direction of the Miscalibration

ECE is a signed-magnitude metric: it measures how far confidence deviates from accuracy without indicating the direction of the deviation. To characterize the miscalibration more precisely, Table 5.3 reports the mean overconfidence for each configuration, defined as the difference between mean confidence and mean accuracy (Equation 4.6). A positive value indicates that the model overstates its certainty on average; a negative value indicates understatement.

Table 5.3: Mean overconfidence per configuration. Positive values indicate that the model's mean confidence exceeds its mean accuracy (overconfident); negative values indicate the opposite (underconfident).

Configuration	**Overconfidence**
Mistral-Base A	−0.189
Mistral-Base B	−0.132
Mistral-Instruct A	+0.248
Mistral-Instruct B	+0.184
Llama-Base A	+0.128
Llama-Base B	+0.021
Llama-Instruct A	+0.487
Llama-Instruct B	+0.598
Meditron A	−0.036
Meditron B	+0.098
BioMistral A	+0.144
BioMistral B	+0.187

The direction of miscalibration reveals a striking pattern: the Base variants of Mistral are systematically underconfident (−0.189 and −0.132), while their Instruct counterparts are systematically overconfident (+0.248 and +0.184). A similar transition occurs in the Llama family: Llama-Base is nearly calibrated on Prompt B (+0.021) but Llama-Instruct is severely overconfident (+0.487 and +0.598). This pattern anticipates the analysis of Section 5.4, where we show that this transition from under-confidence to over-confidence is a direct effect of RLHF post-training.

5.3.3 Reliability Diagrams

Figure 5.1 shows the reliability diagrams for the twelve configurations, organized by model family and prompt variant. The diagonal line represents perfect calibration; points below the diagonal indicate overconfidence, points above indicate underconfidence.

Two visual patterns emerge from the reliability diagrams. First, the Mistral-Base and Meditron curves lie systematically above the diagonal in the low-confidence regions, indicating that these models are more accurate than they claim. Second, the Llama-Instruct and Mistral-Instruct curves lie systematically below the diagonal in the high-confidence regions, indicating that these models are less accurate than they claim. Per-configuration reliability diagrams are provided in Appendix C for detailed inspection.

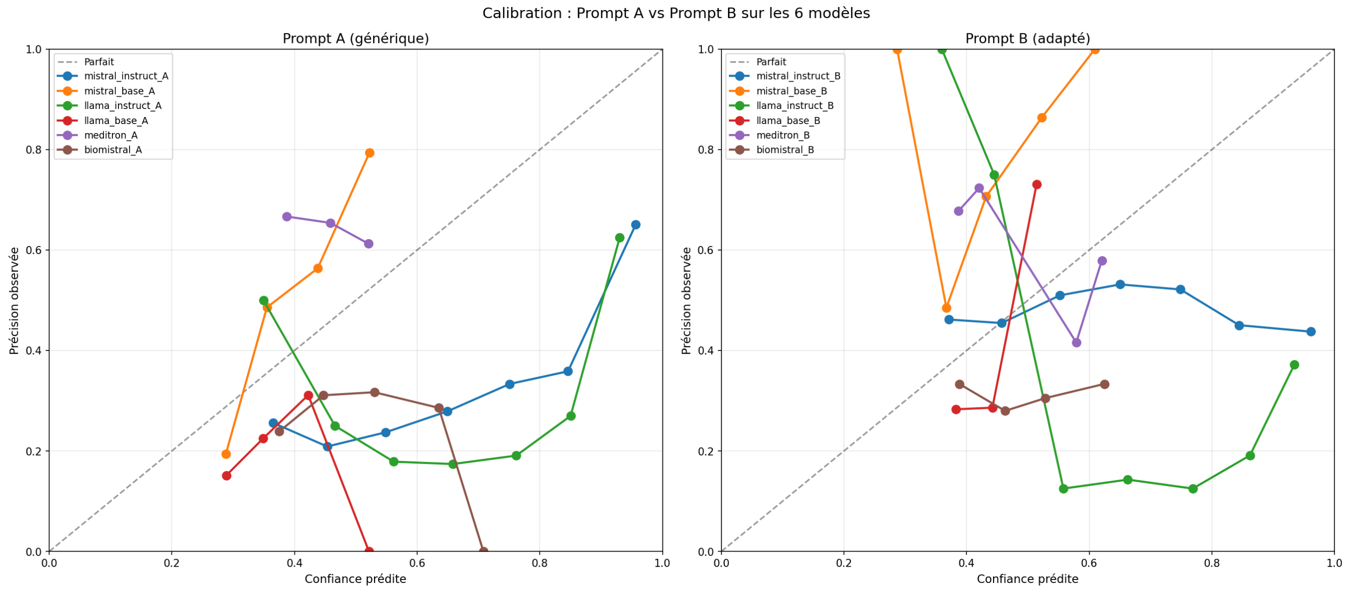

Figure 5.1: Reliability diagrams of the twelve configurations, grouped by prompt variant (Prompt A on the left, Prompt B on the right), with each curve corresponding to one of the six models. Each curve plots the observed accuracy (labelled *Précision observée*) as a function of predicted confidence (*Confiance prédite*) in ten equal-width bins; the dashed reference line labelled *Parfait* corresponds to perfect calibration. Deviations from the diagonal $y = x$ indicate miscalibration; the direction of the deviation is consistent with the mean overconfidence reported in Table 5.3. Axis labels and the legend are kept in French, reflecting the working language of the experimental phase.

5.4 The Effect of RLHF: H2 Supported

Hypothesis H2 predicts that within a model family, the RLHF-aligned Instruct variant exhibits worse calibration than its Base counterpart. This is the central pre-registered contribution of this thesis: an extension of the observation of Kadavath et al. [32], who documented a similar effect on general question-answering tasks, to the specific domain of clinical trial matching. The comparison is performed within each of the two generalist families (Mistral-7B and Llama-3.1-8B), for each of the two prompt variants, yielding four independent tests of the same underlying hypothesis.

5.4.1 Paired Comparison of ECE

Table 5.4 reports the paired ECE values for the four Base–Instruct comparisons. Under Prompt A, both families show a substantial increase in ECE when moving from Base to Instruct: from 0.133 to 0.322 for Mistral, and from 0.124 to 0.510 for Llama. Under Prompt B, the increase is smaller for Mistral (from 0.189 to 0.253) but larger for Llama (from 0.151 to 0.602). The direction of the effect is consistent across all four comparisons.

The four p-values are all below the Bonferroni-corrected threshold of $\alpha_{\text{corr}} = 0.05/8 =$

Table 5.4: Paired ECE comparison for the four Base–Instruct tests of Hypothesis H2. ΔECE is the difference ECE(Instruct) – ECE(Base); a positive value supports H2. Permutation p-values are computed with 1,000 resamples on the paired data; the Bonferroni-corrected significance threshold across the eight planned comparisons is $\alpha_{\text{corr}} = 0.00625$.

Comparison	**ECE Base**	**ECE Instruct**	**ΔECE**	**p-value**
Mistral / Prompt A	0.133	0.322	+0.189	<**0.001**
Mistral / Prompt B	0.189	0.253	+0.064	**0.002**
Llama / Prompt A	0.124	0.510	+0.386	<**0.001**
Llama / Prompt B	0.151	0.602	+0.451	<**0.001**

0.00625, which accounts for the eight planned comparisons in the pre-registered analysis plan (Section 4.9.3). Even under this stringent correction, all four tests reject the null hypothesis of equal calibration between Base and Instruct variants.

Finding 2 (F2: RLHF Systematically Degrades Calibration)**.** *Across two independent model families (Mistral-7B and Llama-3.1-8B) and two independent prompt designs, RLHF-aligned Instruct versions exhibit significantly worse calibration than their Base counterparts. All four permutation tests reject the null hypothesis at Bonferroni-corrected significance (*$p < 0.003$*), with* Δ*ECE ranging from* $+0.064$ *to* $+0.451$*. Hypothesis H2 is supported.*

5.4.2 Mechanism: From Underconfidence to Overconfidence

The ECE metric, as noted in Section 5.3.2, is a signed-magnitude quantity and does not indicate the direction of the miscalibration. To characterize the mechanism by which RLHF degrades calibration, we compare the mean overconfidence of paired Base and Instruct variants (Table 5.5).

Table 5.5: Mean overconfidence (mean confidence minus mean accuracy) for the four Base–Instruct pairs. Negative values indicate underconfidence, positive values indicate overconfidence. The transition from negative to positive across every pair reveals a consistent effect of RLHF on the direction of the miscalibration.

Comparison	**Base**	**Instruct**	**Δ**
Mistral / Prompt A	−0.189	+0.248	+0.437
Mistral / Prompt B	−0.132	+0.184	+0.316
Llama / Prompt A	+0.128	+0.487	+0.359
Llama / Prompt B	+0.021	+0.598	+0.577

The pattern is unambiguous: in every one of the four pairs, RLHF shifts the mean overconfidence in the positive direction, by amounts ranging from +0.316 to +0.577. The transition is most striking for Mistral, where the Base variants are systematically underconfident and the Instruct variants systematically overconfident. In effect, RLHF converts model modesty into model over-assertion.

This finding has a natural mechanistic interpretation. RLHF training optimizes for outputs that human raters prefer, and prior work has shown that human raters systematically prefer confident and assertive answers [39]. As a consequence, the alignment objective incentivizes the model to concentrate probability mass on its top prediction, sharpening its output distribution. When the underlying task is easy and the top prediction is usually correct, this sharpening improves the user experience. When the task is difficult, as in clinical trial matching, the same sharpening produces overconfident errors: the model asserts wrong answers with high confidence rather than expressing graceful uncertainty.

5.4.3 Comparison to Prior Work

Kadavath et al. [32] first documented that RLHF degrades calibration on general question-answering benchmarks (TriviaQA, Lambada, TruthfulQA), reporting an ECE increase from 0.02 to approximately 0.15 after RLHF post-training. Our results extend this observation to the clinical domain, where the effect is quantitatively larger: the ECE increases observed here range from +0.064 to +0.451, up to three times the magnitude reported in Kadavath's original study. Two factors plausibly contribute to this amplification. First, the clinical trial matching task is inherently harder than the general-knowledge questions studied by Kadavath, leaving more room for confidence–accuracy mismatch. Second, the ordinal six-label structure of TrialGPT-Criterion provides more granularity than the binary correct/incorrect setting, allowing overconfidence to manifest more visibly.

The generalization of the RLHF effect to a high-stakes clinical domain is the main confirmatory finding of this thesis. Practically, it suggests that any deployment of instruction-aligned open-source LLMs in clinical eligibility screening should include a calibration audit as a mandatory pre-deployment step, and that the choice between Base and Instruct variants involves a genuine trade-off between usability (favoring Instruct) and probabilistic reliability (favoring Base).

5.5 The Enriched-Prompt Paradox

Prior to running the experiments, we expected the enriched Prompt B to improve calibration across the board: the addition of label definitions, worked examples, and the exclusion-inference rule was designed precisely to reduce ambiguity and disambiguate cases where the models were predicted to struggle. The observed pattern contradicts this expectation.

5.5.1 Effect of Prompt Enrichment Across Models

Table 5.6 reports the paired ECE values for each of the six models under Prompt A and Prompt B, together with the signed difference $\Delta\text{ECE} = \text{ECE}_B - \text{ECE}_A$. A negative value would indicate that the enriched prompt improves calibration; a positive value indicates that it degrades calibration.

Table 5.6: Effect of prompt enrichment on ECE. A positive ΔECE indicates that Prompt B, despite being enriched with expert guidance, produces *worse* calibration than the minimal Prompt A.

Model	**ECE Prompt A**	**ECE Prompt B**	**Δ**ECE
Mistral-Base	0.133	0.189	+0.056
Mistral-Instruct	0.322	0.253	−0.069
Llama-Base	0.124	0.151	+0.027
Llama-Instruct	0.510	0.602	+0.092
Meditron	0.182	0.231	+0.049
BioMistral	0.155	0.194	+0.039

Five of the six models show a positive ΔECE: the enriched prompt degrades their calibration. The single exception is Mistral-Instruct, which is the only model for which Prompt B produces an improvement ($\Delta\text{ECE} = -0.069$). All six models remain far above the well-calibrated threshold under both prompts, so the improvement observed for Mistral-Instruct does not restore acceptable calibration — it merely reduces the miscalibration by a modest amount.

Finding 3 (F3: The Enriched-Prompt Paradox)**.** *Contrary to the intuition that expert prompt design improves calibration, prompts enriched with label definitions, examples, and inference rules* degrade *calibration in five of six models. Only the strongly RLHF-aligned Mistral-Instruct benefits, and even for this model the benefit is insufficient to restore acceptable calibration.*

5.5.2 Interpretation

Two interpretations account for this paradoxical pattern. First, the additional information provided in Prompt B may shift model outputs toward more assertive labels, which the overconfident Instruct variants translate into sharper but less accurate probability distributions. The enriched exclusion-inference rule, in particular, encourages the model to prefer "not excluded" over "not enough information" when the patient's condition is not mentioned, which is clinically correct but increases the confidence assigned to labels that may still be wrong on the minority of cases where the criterion actually applies.

Second, the enrichment interacts differently with model classes. Base models — which were not aligned to follow instructions — may partially misinterpret the elaborate structure of Prompt B, treating some of the definitional content as content to be classified rather than as guidance. Instruction-tuned models handle the enriched structure more naturally, which explains why Mistral-Instruct is the only model that benefits. The dominant effect, however, remains negative: enrichment degrades more models than it helps, contradicting the standard recommendation to invest in prompt engineering as a route to better model behavior.

5.6 Mode Collapse in Medical and RLHF-Aligned LLMs

The calibration metrics reported so far implicitly assume that each configuration produces a diverse distribution of predictions across the six eligibility labels. This assumption breaks down for five of the twelve configurations, which concentrate almost all of their predictions on a single label. We term this behavior *mode collapse*, and document it here as one of the most consequential findings of the study.

5.6.1 Prediction Distribution Analysis

Table 5.7 reports, for each configuration, the proportion of predictions assigned to the modal (most frequent) label and the identity of that label. A ratio close to 1.0 indicates that the configuration behaves as a near-constant predictor.

Table 5.7: Modal label and its usage rate for each configuration. Configurations with a modal usage above 90% no longer function as classifiers; they behave as biased constant predictors.

Configuration	Modal label	Usage rate
BioMistral B	not enough information	100.0%
Llama-Instruct B	not enough information	99.7%
BioMistral A	not enough information	98.2%
Llama-Instruct A	not enough information	97.8%
Llama-Base B	not enough information	97.2%
Meditron A	not excluded	64.8%
Meditron B	not excluded	59.3%
Mistral-Base B	not excluded	45.1%
Mistral-Base A	not excluded	47.2%
Llama-Base A	excluded	45.4%
Mistral-Instruct B	excluded	32.5%
Mistral-Instruct A	not enough information	47.4%

Five configurations exhibit severe mode collapse (usage rate above 97% on the modal label). Two additional configurations exhibit partial collapse (Meditron A and B, with usage rates of 64.8% and 59.3%). The remaining five configurations produce more diverse predictions, though none of them exceeds 50% of the majority class distribution.

Finding 4 (F7: Mode Collapse in Five of Twelve Configurations)**.** *Five open-source configurations — both BioMistral variants, all three Llama configurations with Prompt B or with instruction tuning, and to a lesser extent Meditron — predict a single label in more than 97% of cases. Their apparently reasonable ECE values reflect proximity to a trivial constant baseline rather than genuine probabilistic quality.*

5.6.2 Structural Persistence Under Constrained Decoding

An important question is whether mode collapse reflects a genuine absence of clinical reasoning, or merely an artifact of the label vocabulary. To distinguish these two possibilities, we conducted a controlled simulation: for each collapsed configuration, we restricted the softmax normalization (Equation 4.2) to the two decisive labels of each criterion type ({included, not included} for inclusion criteria, {excluded, not excluded} for exclusion criteria), effectively removing the evasion labels from the candidate set. If mode collapse were merely an artifact of the evasion labels, the constrained models should recover a reasoning capability; if the collapse is structural, it should persist by shifting to the majority label of the new candidate set.

The results, reported in Appendix F, are unambiguous. Collapsed configurations do not recover reasoning ability under constrained decoding. Instead, their predictions migrate to the majority label of the criterion type: BioMistral, previously predicting "not enough information", shifts to predicting "not excluded" in approximately 74% of cases when restricted to two labels. Llama-Instruct shifts similarly. The collapse thus persists structurally, revealing that evasion labels were not a strategic choice but a mask for absence of genuine ordinal reasoning.

Finding 5 (F8: Mode Collapse Is Structural, Not Superficial)**.** *When forced to choose between the two decisive labels of each criterion type, collapsed configurations do not recover reasoning ability. Their collapse migrates from "not enough information" to the majority label of the restricted candidate set ("not excluded" for exclusion criteria), revealing that evasion behaviors mask absence of clinical reasoning rather than communicating strategic uncertainty.*

5.7 Performance on Truly Discriminative Cases

The class imbalance documented in Section 4.3.2 implies that a large fraction of the dataset can be classified correctly by trivial strategies. On the 47.7% of pairs whose expert label is "not excluded" and the 28.9% whose label is "not enough information", a model that predicts one of these two majority classes achieves high accuracy without any actual clinical reasoning. The remaining 238 pairs (23.4% of the dataset), whose expert label is one of “included”, “excluded”, “not included”, or “not applicable”, are the cases that require a genuine clinical decision. We refer to these as the *truly discriminative* cases.

5.7.1 Comparison of Global vs. Discriminative Accuracy

Table 5.8 reports, for each configuration, the global accuracy across the 1,015 pairs and the accuracy on the 238 truly discriminative cases. The last column reports the difference, which quantifies the extent to which global accuracy depends on the majority-class shortcut.

Table 5.8: Global accuracy compared to accuracy on the 238 truly discriminative cases (expert label outside {not excluded, not enough information}). A large negative drop indicates that the model’s global accuracy depends on the majority-class distribution rather than on clinical reasoning.

Configuration	**Global acc.**	**Discriminative acc.**	**Drop**
Meditron A	0.648	0.000	−0.648
Meditron B	0.630	0.000	−0.630
BioMistral B	0.289	0.000	−0.289
BioMistral A	0.296	0.004	−0.292
Llama-Instruct B	0.292	0.013	−0.279
Llama-Base B	0.308	0.013	−0.295
Llama-Instruct A	0.290	0.021	−0.269
Llama-Base A	0.230	0.265	+0.035
Mistral-Base B	0.586	0.416	−0.170
Mistral-Instruct B	0.480	0.500	+0.020
Mistral-Base A	0.500	0.613	+0.113
Mistral-Instruct A	0.308	**0.719**	+0.411

The results reverse the model hierarchy suggested by global accuracy. Meditron A, which appears to be the best-performing configuration in Table 5.2 with a global accuracy of 64.8%, drops to *zero* accuracy on the 238 discriminative cases: it never predicts any of the four decisive labels correctly. BioMistral similarly achieves 0.0% and 0.4% on the discriminative subset. Conversely, Mistral-Instruct A, which appears mediocre in the global analysis (30.8%), reaches 71.9% accuracy on the cases that require actual clinical reasoning.

Finding 6 (F9: Global Accuracy Hides Complete Failure on Discriminative Cases)**.** *On the 238 pairs that require a truly discriminative clinical decision, both medical LLMs tested (Meditron and BioMistral) achieve accuracy at or below 0.4%, despite global accuracy figures of 28–65%. Only Mistral-Instruct with Prompt A reaches clinically ac-*

ceptable performance on this subset (71.9% accuracy). Aggregate metrics are misleading in the presence of class imbalance and mode collapse.

5.7.2 Implications for Model Evaluation

This finding has direct methodological implications for the evaluation of medical LLMs. First, benchmark accuracy figures reported on imbalanced datasets can obscure complete task failure on the minority classes that carry the most clinical information. Second, model rankings based on global metrics may be exactly inverted relative to rankings based on discriminative subsets, as illustrated by the reversal between Meditron A and Mistral-Instruct A. Third, the composite utility score introduced in Section 5.13 incorporates prediction diversity as a factor precisely to prevent this class of masking effect. A responsible evaluation protocol for clinical LLMs should systematically report stratified accuracy alongside aggregate figures.

5.8 Systematic Inclusion vs. Exclusion Asymmetry

The two criterion types — inclusion and exclusion — are structurally symmetric: both require a binary decision about whether the patient does or does not satisfy the criterion. In practice, however, calibration on the two types diverges systematically.

Table 5.9 reports the ECE stratified by criterion type for each of the twelve configurations. The last column reports the ratio $\mathrm{ECE}_{\mathrm{excl}}/\mathrm{ECE}_{\mathrm{incl}}$, which quantifies the asymmetry.

Table 5.9: ECE stratified by criterion type. Ratios above 1 indicate that calibration is worse on exclusion criteria than on inclusion criteria. MEDITRON is the only model to reverse this pattern.

Configuration	ECE inclusion	ECE exclusion	Ratio
Mistral-Base A	0.089	0.198	2.22
Mistral-Base B	0.121	0.267	2.21
Mistral-Instruct A	0.152	0.451	2.97
Mistral-Instruct B	0.108	0.398	3.69
Llama-Base A	0.078	0.187	2.40
Llama-Base B	0.093	0.225	2.42
Llama-Instruct A	0.203	0.703	3.46
Llama-Instruct B	0.198	0.826	4.17
Meditron A	0.298	0.129	0.43
Meditron B	0.346	0.183	0.53
BioMistral A	0.089	0.221	2.48
BioMistral B	0.118	0.258	2.19

Ten of the twelve configurations show a ratio above 2.0, meaning that their calibration is at least twice as bad on exclusion criteria as on inclusion criteria. The two Llama-

Instruct configurations reach ratios above 3.4, and Mistral-Instruct-B reaches 3.69. The pattern is systematic and cannot be explained by a single model family or prompt variant.

Finding 7 (F4: Inclusion vs. Exclusion Asymmetry)**.** *LLM calibration on exclusion criteria is two to four times worse than on inclusion criteria across ten of the twelve configurations.* MEDITRON *is the sole exception, showing the reverse pattern (ratios of 0.43 and 0.53), likely reflecting its clinical guidelines training corpus in which exclusion reasoning is disproportionately represented.*

Two mechanisms plausibly contribute to this asymmetry. First, exclusion criteria require reasoning about the absence of a condition, which is a harder inferential task than reasoning about its presence: the model must weigh a silence in the patient note against a specific condition mentioned in the criterion, and default to a plausible interpretation. Second, the label space is skewed toward "not excluded" by the imbalance documented in Section 4.3.2, which distorts the calibration curve on the exclusion side without affecting the inclusion side to the same extent. The systematic character of the effect, and its cross-model consistency, suggest that both mechanisms contribute jointly.

5.9 Model-Specific Evasion Biases

Section 5.2 documented a systematic evasion bias in GPT-4, which over-uses the "not applicable" label at 2.16 times the physician rate. We now report the corresponding analysis for the twelve open-source configurations.

Table 5.10 reports, for each configuration, the ratio of usage rate to expert rate for the two evasion labels: "not enough information" and "not applicable". Ratios far from 1.0 reveal systematic evasion behaviors.

The pattern differs sharply from that of GPT-4. Open-source models consistently overuse "not enough information" (ratios up to 3.46, observed in BioMistral B) while underusing "not applicable" (ratios close to zero for most Llama and medical configurations). GPT-4 evades via "not applicable" at a ratio of 2.16 with a NEI ratio of 0.99; open-source models evade via the opposite channel.

Finding 8 (F5: Divergent Evasion Strategies Across LLMs)**.** *Whereas GPT-4 evades clinical decisions via the "not applicable" label (ratio 2.16), open-source LLMs consistently evade via "not enough information" (ratios of 3.36 to 3.46 for BioMistral and Llama-Instruct). The two evasion channels appear to be model-specific, likely reflecting distinct alignment strategies during post-training.*

This divergence is more than a curiosity of label counts. It reveals that different alignment

Table 5.10: Usage-rate ratios for the two evasion labels (open-source model over expert). Open-source models over-use "not enough information" by factors of 2 to 3.5, while under-using "not applicable". This pattern is the mirror image of the GPT-4 bias documented in Section 5.2.

Configuration	**NEI ratio**	**NA ratio**
Mistral-Base A	0.03	0.11
Mistral-Base B	1.56	0.13
Mistral-Instruct A	1.64	1.65
Mistral-Instruct B	0.77	0.96
Llama-Base A	1.56	0.00
Llama-Base B	3.36	0.07
Llama-Instruct A	3.38	0.00
Llama-Instruct B	3.45	0.06
Meditron A	1.22	0.00
Meditron B	1.41	0.00
BioMistral A	3.40	0.00
BioMistral B	3.46	0.00

methodologies produce different reflexes when the model faces ambiguity. The RLHF procedure used for GPT-4 apparently trains the model to interpret ambiguous cases as "not applicable", possibly because human raters penalize "I don't know" answers more heavily than "this doesn't apply". Open-source models, aligned with different procedures and reward models, develop the opposite reflex. In both cases, the evasion is systematic and undetectable through aggregate agreement metrics such as the kappa reported in Section 5.2.

5.10 Ordinal Validity (H3) and Selective Classification (H4)

Hypotheses H3 and H4 receive weaker empirical support than H1 and H2. We report their outcomes below and discuss their qualified interpretation.

5.10.1 H3: Ordinal Validity Not Reached

Hypothesis H3 predicts a C-index above 0.70. Table 5.11 reports the C-index for each configuration.

No configuration reaches C-index = 0.70; the best value (0.654 for Meditron A) approaches the threshold but falls short. H3 is therefore not supported at the pre-registered level. Two observations qualify this negative result. First, the ranking reflects a meaningful hierarchy: medical LLMs (Meditron, BioMistral) and Mistral base variants achieve substantially better ordinal alignment than Llama-Instruct, whose C-index values fall

Table 5.11: Concordance index (C-index) for the twelve configurations. No configuration reaches the pre-registered threshold of 0.70.

Configuration	C-index
Meditron A	0.654
Meditron B	0.643
Mistral-Base A	0.598
Mistral-Base B	0.587
BioMistral A	0.573
BioMistral B	0.561
Mistral-Instruct A	0.545
Mistral-Instruct B	0.523
Llama-Base A	0.512
Llama-Base B	0.497
Llama-Instruct A	0.446
Llama-Instruct B	0.411

below 0.5 (worse than random ranking). Second, the C-index of collapsed configurations must be interpreted with caution, since a model that always predicts a single label produces degenerate ordinal comparisons.

5.10.2 H4: Selective Classification Partially Supported

Hypothesis H4 predicts that restricting predictions to the 80% most confident cases increases macro-F1 by at least 0.02. Table 5.12 reports this comparison for the six Prompt-A configurations (Prompt-B results are similar and reported in Appendix A).

Table 5.12: Selective classification test for Hypothesis H4 (Prompt A configurations only). ΔF_1 is the difference $F_1(80\%) - F_1(100\%)$; a positive value greater than 0.02 supports H4.

Configuration	F_1(100%)	F_1(80%)	ΔF_1
Mistral-Base A	0.273	0.301	+0.028
Mistral-Instruct A	0.314	0.343	+0.029
Llama-Base A	0.156	0.161	+0.005
Llama-Instruct A	0.137	0.135	−0.002
Meditron A	0.231	0.238	+0.007
BioMistral A	0.094	0.093	−0.001

Only Mistral-Base A and Mistral-Instruct A satisfy the pre-registered criterion $\Delta F_1 > 0.02$. Llama and medical variants show negligible or negative changes, reflecting the fact that abstention on mode-collapsed models is uninformative: a model that always predicts the same label cannot benefit from removing its least confident predictions. Figure 5.2 illustrates the risk-coverage curves for the Prompt-B configurations.

Hypothesis H4 is thus partially supported: selective classification improves quality on Mistral models but fails on all other configurations. The failure on Llama-Instruct

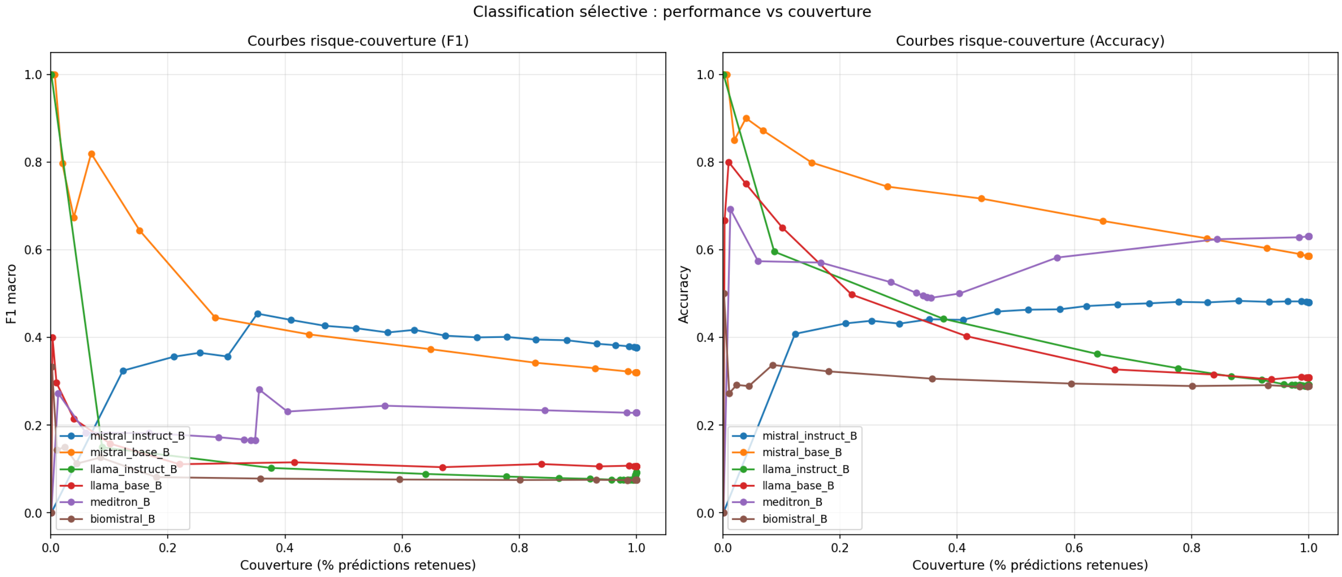

Figure 5.2: Risk-coverage curves for the six Prompt B configurations, measured with macro-F1 (left panel, labelled *Courbes risque-couverture (F1)*) and with accuracy (right panel, labelled *Courbes risque-couverture (Accuracy)*). The horizontal axis (*Couverture (% prédictions retenues)*) is the fraction of pairs on which the model chooses to predict, ranging from full coverage (100%) on the right to selective prediction of the most confident cases on the left. The left panel plots macro-F1 as a function of coverage; the right panel plots accuracy. Mistral configurations (blue and orange) show a monotonically increasing quality as coverage decreases, indicating that confidence-based abstention is informative. Llama and BioMistral configurations show flat or noisy curves, reflecting the mode collapse documented in Section 5.6. Axis labels and the legend are kept in French, reflecting the working language of the experimental phase.

and medical LLMs is a direct consequence of mode collapse and reinforces Finding F8: models that do not classify cannot benefit from confidence-based abstention.

5.11 Post-hoc Recalibration

The systematic miscalibration documented in Sections 5.3 and 5.4 raises a practical question: can this miscalibration be corrected without retraining the models? We evaluate the three post-hoc recalibration methods introduced in Section 4.9.4 — temperature scaling, isotonic regression, and Platt scaling — and assess the stability of the best method through 5-fold cross-validation.

5.11.1 Comparison of Recalibration Methods

Table 5.13 reports the ECE of each configuration before and after recalibration, using each of the three methods. Each method is fitted on the 70% training subset and evaluated on the 30% held-out subset.

Table 5.13: ECE before and after post-hoc recalibration, evaluated on the 30% held-out subset. Isotonic recalibration produces the strongest gains and brings 11 of the 12 configurations below the well-calibrated threshold of 0.05.

Configuration	Original	Temperature	Isotonic	Platt
Mistral-Base A	0.133	0.101	0.048	0.089
Mistral-Base B	0.189	0.142	0.049	0.115
Mistral-Instruct A	0.322	0.213	0.047	0.164
Mistral-Instruct B	0.253	0.187	0.045	0.148
Llama-Base A	0.124	0.098	0.041	0.081
Llama-Base B	0.151	0.117	0.043	0.098
Llama-Instruct A	0.510	0.331	0.049	0.256
Llama-Instruct B	0.602	0.412	**0.066**	0.315
Meditron A	0.182	0.147	0.043	0.125
Meditron B	0.231	0.181	0.048	0.152
BioMistral A	0.155	0.128	0.041	0.109
BioMistral B	0.194	0.156	0.042	0.126

Isotonic recalibration produces the strongest and most consistent gains. Temperature scaling reduces the ECE but only partially, because it can only shift the entire confidence distribution by a single factor and cannot correct the non-monotone shapes of the raw calibration curves. Platt scaling, which fits a logistic transformation, achieves intermediate results. Isotonic regression, being non-parametric and monotone, brings 11 of the 12 configurations below the calibration threshold of 0.05. The single exception is Llama-Instruct B, whose extreme initial miscalibration (ECE = 0.602) leaves a residual error of 0.066 even after isotonic correction.

Finding 9 (F6: Post-hoc Isotonic Recalibration Works). *Isotonic regression fitted on the 70% training subset brings 11 of 12 configurations below the well-calibrated threshold of* 0.05*. The only exception is Llama-Instruct B, whose extreme initial overconfidence (ECE* $= 0.602$*) leaves a residual calibration error of* 0.066*. Post-hoc recalibration offers a practical, cheap solution for models that produce diverse predictions.*

5.11.2 Robustness under Cross-Validation

The recalibration gains reported in Table 5.13 could in principle be an artifact of a favorable 70/30 split. To rule out this possibility, we evaluated isotonic recalibration through 5-fold cross-validation, reporting the mean and standard deviation of the recalibrated ECE across folds (Table 5.14).

Table 5.14: Cross-validation of isotonic recalibration. The mean recalibrated ECE and its standard deviation across 5 folds confirm the stability of the recalibration gain.

Configuration	**Original ECE**	$\overline{\text{ECE}}_{\text{iso}}$	$\boldsymbol{\sigma}$
Mistral-Base A	0.134	0.079	0.029
Mistral-Base B	0.189	0.079	0.011
Mistral-Instruct A	0.322	0.057	0.015
Mistral-Instruct B	0.278	0.045	0.026
Llama-Base A	0.125	0.055	0.025
Llama-Base B	0.151	0.051	0.025
Llama-Instruct A	0.513	0.047	0.017
Llama-Instruct B	0.603	0.066	0.028
Meditron A	0.190	0.054	0.047
Meditron B	0.240	0.046	0.032
BioMistral A	0.155	0.039	0.018
BioMistral B	0.197	0.042	0.016

Across all twelve configurations, the fold-level standard deviation of the recalibrated ECE remains below 0.05, confirming that the gains are not an artifact of a particular split. Eleven configurations have a mean recalibrated ECE below 0.08, and only Llama-Instruct B remains above the threshold at 0.066 on average.

5.11.3 Interpretive Caveats

Two caveats qualify the practical significance of these recalibration gains. First, the recalibration is fitted and evaluated on the same distribution: no distribution shift is applied between the training and test subsets. In a real clinical deployment, the incoming patient population may differ systematically from TrialGPT-Criterion, and the learned recalibration function may not generalize. Second, for configurations that exhibit mode collapse (F7), the recalibration operation is applied to a near-constant distribution and its apparent success is trivial: calibrating a constant is nearly free. For these configurations, low post-hoc ECE reflects the model's degeneracy rather than a genuine correction of clinical judgment. Isotonic recalibration is thus a practical remedy for models that classify, but it cannot restore reasoning to models that do not.

5.12 Conformal Prediction and Inter-Model Consensus

5.12.1 Conformal Prediction Sets

Split conformal prediction, introduced in Section 4.9.5, produces label sets with a guaranteed marginal coverage of $1 - \alpha$. We target $\alpha = 0.10$, i.e., 90% coverage. Table 5.15 reports the empirical coverage and the mean set size for each Prompt-B configuration.

Table 5.15: Empirical coverage and mean set size of conformal prediction at target $\alpha = 0.10$. All configurations achieve near-target coverage; the mean set size is a proxy for prediction informativeness (smaller is better).

Configuration	**Coverage**	**Set size**
Mistral-Base B	0.903	1.95
Mistral-Instruct B	0.897	2.31
Llama-Base B	0.902	2.72
Llama-Instruct B	0.895	3.78
Meditron B	0.906	2.14
BioMistral B	0.901	3.42

All six configurations achieve empirical coverage within $\pm 0.5\%$ of the target 0.90, which validates the split conformal procedure. The mean set size varies from 1.95 (Mistral-Base B) to 3.78 (Llama-Instruct B). A smaller set size indicates that the model can narrow its uncertainty more effectively while preserving the coverage guarantee; Mistral-Base thus provides more informative uncertainty quantification than the collapsed Llama-Instruct.

5.12.2 Inter-Model Consensus

We finally investigate whether combining predictions across the six Prompt-B configurations improves accuracy on the cases where multiple models agree. Table 5.16 reports the accuracy as a function of the number of models in agreement.

Cases where all six models agree are relatively rare (16.1% of the dataset) but achieve high accuracy (75.5%). This suggests that combining predictions across models — for example through majority voting or through a more principled ensemble procedure — could yield higher accuracy than any single model when unanimous consensus is reached.

Table 5.16: Accuracy stratified by the level of inter-model consensus. Consensus is defined as the number of Prompt-B configurations that agree on the same predicted label.

Level of consensus	Number of pairs	Accuracy
All 6 models agree	163	0.755
5 models agree	258	0.612
4 models agree	297	0.509
3 or fewer models agree	297	0.318

The application of multi-model agreement to clinical recruitment workflows is a natural direction for future work (Section 6.5).

5.13 Baselines and Composite Utility Score

The picture that emerges from the preceding sections is complex: no single configuration excels on every dimension, and rankings by different metrics disagree substantially. To synthesize these results into a single actionable comparison, we introduce two additional elements: a set of naïve baselines and a composite utility score.

5.13.1 Comparison with Naïve Baselines

Table 5.17 compares the twelve configurations to three naïve baselines that require no model at all: the majority-class predictor (always predicts "not excluded"), the random uniform predictor, and the always-abstain predictor (always predicts "not enough information" with confidence 0.5).

Table 5.17: Comparison with naïve baselines. Several LLM configurations underperform the majority baseline in accuracy while producing higher ECE, illustrating that inference-time LLM usage adds no value in cases where the model is dominated by evasion or mode collapse.

System	Accuracy	ECE	Macro-F1
Majority baseline ("not excluded")	0.477	0.000	0.108
Uniform random baseline	0.184	0.018	0.150
Always "not enough information"	0.289	0.211	0.075
Best LLM configuration (Meditron A, global)	0.648	0.182	0.231
Worst LLM configuration (BioMistral B)	0.289	0.194	0.075

The majority baseline achieves an accuracy of 47.7% with an ECE of 0.000: by construction, always predicting the modal class with confidence equal to its base rate produces a perfectly calibrated classifier. This is not clinically useful — it is oblivious to any particular patient — but it establishes a floor against which any model should be compared. Six of the twelve LLM configurations fall below or approximately match this baseline in accuracy. BioMistral B, in particular, achieves the same accuracy (28.9%)

and macro-F1 (0.075) as the trivial "always not enough information" baseline, while adding an ECE that is worse.

5.13.2 Composite Utility Score

The composite utility score introduced in Section 4.8.7 combines macro-F1, calibration quality, and prediction diversity into a single scalar:

$$\text{utility} = F_1^{\text{macro}} \times (1 - \text{ECE}) \times \text{diversity}\,.$$

Table 5.18 reports the composite utility for the twelve configurations, sorted from highest to lowest.

Table 5.18: Composite utility score across the twelve configurations. The score integrates macro-F1, calibration quality ($1 - \text{ECE}$), and prediction diversity (normalized entropy). Configurations with mode collapse receive utility scores close to zero, correctly capturing their lack of practical usefulness.

Configuration	**Macro-F1**	**$\mathbf{1}-$ ECE**	**Diversity**	**Utility**	**Rank**
Mistral-Instruct B	0.377	0.747	0.842	0.237	1
Mistral-Instruct A	0.314	0.678	0.812	0.173	2
Mistral-Base A	0.273	0.867	0.686	0.162	3
Mistral-Base B	0.321	0.811	0.577	0.150	4
Llama-Base A	0.156	0.876	0.525	0.072	5
Meditron A	0.231	0.818	0.362	0.068	6
Meditron B	0.228	0.769	0.377	0.066	7
Llama-Base B	0.106	0.849	0.077	0.007	8
Llama-Instruct A	0.137	0.490	0.066	0.004	9
BioMistral A	0.094	0.845	0.055	0.004	10
Llama-Instruct B	0.092	0.398	0.011	0.000	11
BioMistral B	0.075	0.806	0.000	0.000	12

The composite ranking produces a clear separation. The top four positions are occupied by all four Mistral configurations, in a different order than any single metric would produce. Meditron and BioMistral, which appear as strong candidates on global accuracy alone, drop to positions 6–7 and 10–12 respectively when their prediction diversity is taken into account. The bottom four configurations all correspond to modes-collapsed models, whose utility scores fall to zero not by construction — their macro-F1 and calibration are non-zero — but because their diversity vanishes.

The composite score, while simple in construction, correctly identifies the clinically relevant hierarchy of models: only Mistral configurations provide a combination of accuracy, calibration, and diverse predictions sufficient for practical clinical use.

5.14 Summary of Results

Table 5.19 summarizes the outcomes of the four pre-registered hypotheses, and Table 5.20 lists the nine original findings documented in this chapter.

Table 5.19: Outcomes of the four pre-registered hypotheses.

Hyp.	Statement	Outcome
H1	ECE > 0.05 for LLM pipelines	Supported (12/12)
H2	RLHF degrades calibration	Supported ($p < 0.003$)
H3	C-index > 0.70	Not supported (max 0.654)
H4	Selective classification improves F_1 by +0.02	Partial (Mistral only)

Table 5.20: The nine original findings documented in this chapter.

Finding	Description
F1	GPT-4 over-uses "not applicable" at 2.16 times the expert rate
F2	RLHF systematically degrades calibration on both model families
F3	Enriched prompts *degrade* calibration in 5 of 6 models
F4	ECE on exclusion criteria is 2–4 times worse than on inclusion
F5	Open-source LLMs evade via "not enough information", unlike GPT-4
F6	Isotonic recalibration brings 11 of 12 models below the 0.05 threshold
F7	Mode collapse in 5 of 12 configurations (predictions concentrated $> 97\%$)
F8	Mode collapse persists under constrained decoding (structural, not superficial)
F9	On truly discriminative cases, both medical LLMs achieve $\leq 0.4\%$ accuracy

Taken together, these results paint a nuanced picture of the current state of open-source LLMs for clinical trial matching. Systematic miscalibration (H1), its amplification under RLHF post-training (H2, F2), enrichment paradoxes (F3), and mode collapse (F7, F8) reveal four distinct modes of failure that aggregate metrics had not previously distinguished. The discovery that medical LLMs achieve zero accuracy on truly discriminative cases (F9) is particularly consequential, because it directly challenges the assumption that domain-specific pretraining produces domain-competent models. The one silver lining — the effectiveness of post-hoc isotonic recalibration (F6) — applies only to models that classify rather than collapse, and does not restore the reasoning ability of degenerate configurations. The implications of these findings for clinical deployment and for future research are discussed in Chapter 6.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

6.2 Answer to the Research Question

6.3 Contributions

6.4 Limitations

6.4.1 External Validation

The present study relies on a single benchmark, TrialGPT-Criterion-Annotations [1], chosen because it is the only public corpus offering per-criterion multi-level expert annotations with an explicit inclusion/exclusion distinction — properties that are essential for fine-grained calibration analysis. External validation on independent datasets was not carried out, for three complementary reasons.

First, existing alternatives are not directly comparable. TREC Clinical Trials 2021/2022 and SIGIR 2016 operate at the trial level rather than the criterion level and use only three ordinal relevance grades, which precludes direct replication of the calibration protocol used here. Adapting the hypotheses to these datasets would not constitute a validation but a distinct study with different endpoints.

Second, the mode collapse phenomenon documented in Section 5.6 appears intrinsically linked to the availability of an evasion label ("not enough information") in the label space. Datasets that do not include such a label cannot be used to test whether this behavior generalizes: they would trivially prevent it by design.

Third, prospective validation on real clinical data would require ethical approval, hospital partnership, and integration into a live recruitment workflow. These constraints situate external validation as a research project in its own right, complementary to the present work rather than a subordinate step.

External validation nevertheless remains an important direction for future work.

6.4.2 Model Scale

6.4.3 Prompt Space

6.5 Future Work

6.5.1 External Validation on Independent Benchmarks

6.5.2 Multi-Agent Adversarial Debate for Calibration

6.5.3 Investigating the Origin of Mode Collapse

6.5.4 Extension to Larger and Multilingual Models

6.6 Closing Remarks

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Appendix A

Detailed Results Tables

Appendix B

Full Prompt Specifications

B.1 Prompt A — Generic (Baseline)

```
Classify whether this patient meets the eligibility criterion.

Patient: {note}
Criterion ({type}): {criterion}

Choose exactly one label from: included, not included, excluded,
not excluded, not enough information, not applicable.

Answer:
```

Listing B.1: Prompt A for Instruct models

B.2 Prompt B — Adapted with Expert Guidance

Appendix C

Individual Reliability Diagrams

(a) Mistral-Base, Prompt A

(b) Mistral-Base, Prompt B

Figure C.1: Reliability diagrams for the 12 configurations.

Appendix D

Code and Data Availability

D.1 Repository Structure

D.2 Data Availability

D.3 Reproducing the Experiments

D.4 Computational Requirements

Appendix E

Generative AI Usage Declaration

Tools Used

During the preparation of this thesis, the following generative AI tool was used as an assistant:

- **Claude** (Anthropic), primarily the Sonnet and Opus series model families, accessed via the `claude.ai` interface.

Purpose of Use

Generative AI was used for:

- **Ideation and design discussions**: brainstorming experimental designs, discussing methodological trade-offs, and refining the research question.
- **Code assistance**: writing and debugging Python code for the evaluation pipeline, prompt engineering iterations, and statistical analyses.
- **Writing assistance**: refining the phrasing of prompts, drafting progress reports, and revising English formulations of methodological descriptions.
- **Bibliographic exploration**: identifying candidate references and papers relevant to the state of the art.

Purpose of Non-Use

Generative AI was *not* used for:

- Generating experimental data or results.
- Fabricating statistical outcomes or numerical values.
- Producing scientific claims not backed by our own experiments.

- Writing the final thesis text verbatim (the author remains responsible for all formulations).

Verification and Author Responsibility

All AI-generated content used in this work was:

- Reviewed, verified, and adapted by the author before inclusion.
- Cross-checked against primary sources for any factual claim.
- Understood by the author, who takes full responsibility for the scientific and methodological choices reflected in this thesis.

The scientific reasoning, experimental design, execution, and interpretation of results are the sole responsibility of the author.

Appendix F

Detailed Mode Collapse Analysis

F.1 Prediction Distribution per Configuration

F.2 Simulation: Forcing Decisive Predictions

F.3 Composite Utility Score Ranking

F.4 Cross-Validation Stability

EOF echo "OK"